Overview of thorax

By

Dr.Afsheen Khan

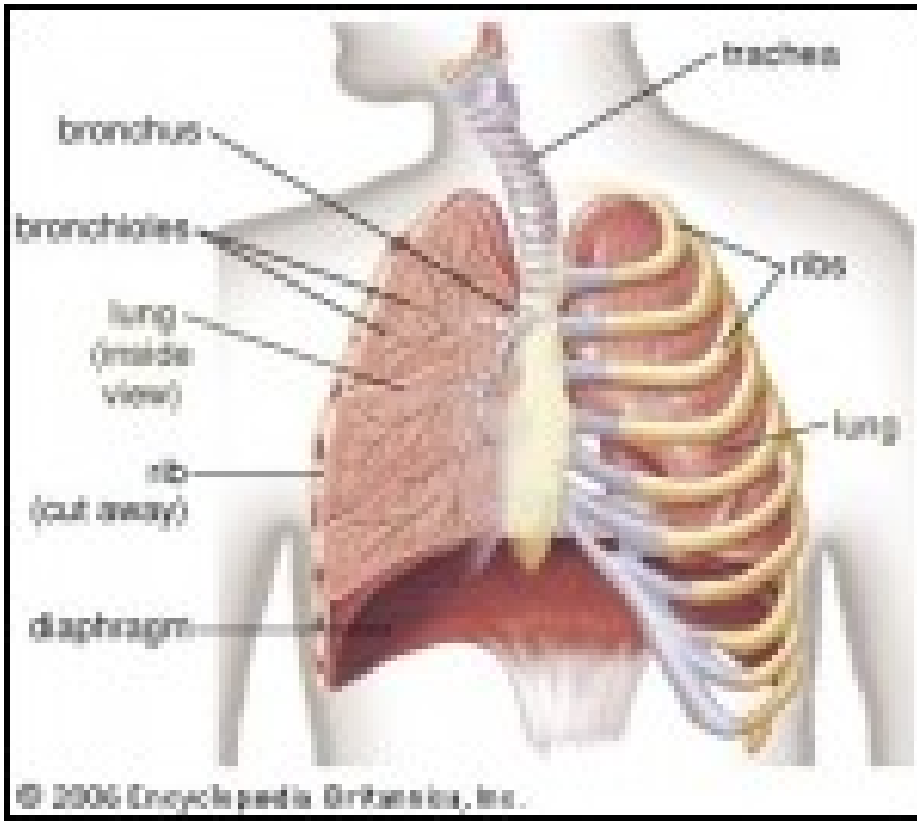
objectives

- At the end of the class students should able to understands

1. Structural out line of the thorax.
2. Different components of the thoracic cage.
3. Different compartments of the thoracic cavity.
4. Boundaries of the thoracic inlet & outlet

introduction

1. Region of trunk b/w neck & abdomen.
2. Separated from abdomen by diaphragm.
3. Skeletal framework of thoracic cage.
4. Cavity contain cordio respiratory viscera
5. Provide protection to important viscera.



Over view of thorax

1. Body wall.
    1. Skin.
    2. Superficial fascia.
2. blood supply by
    1. Intercostal.
    2. Subcostal.
    3. Internal thoracic.
3. Lymphatic drainage
    1. Axillary nodes.

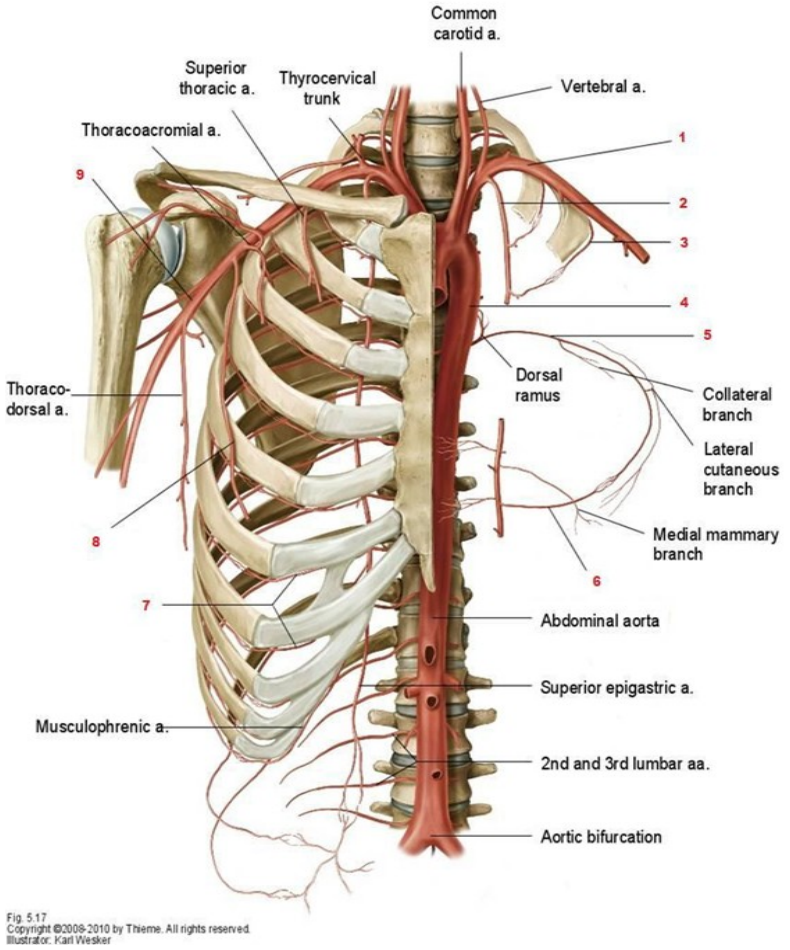
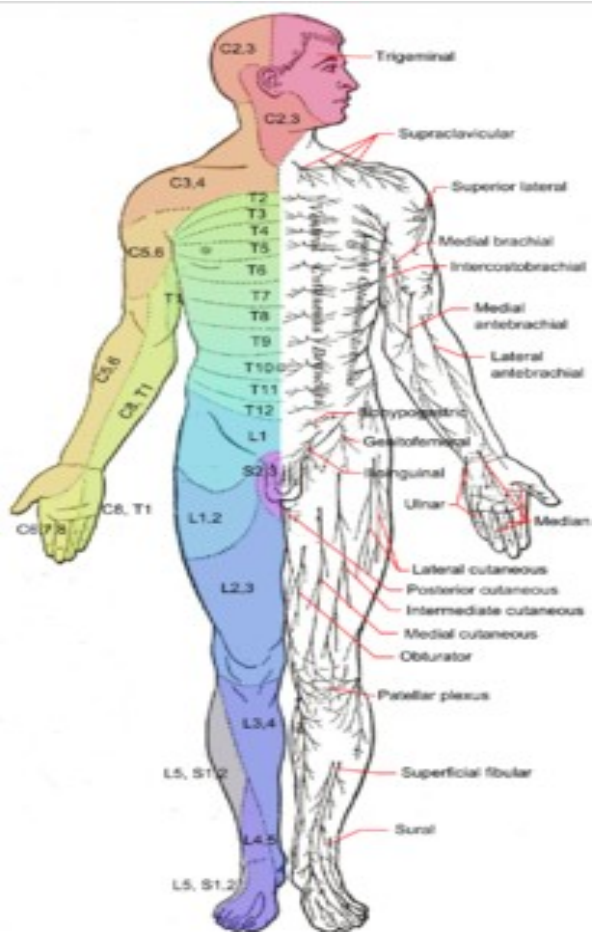


Fig. 5.17
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Illustrator: Karl Wesker

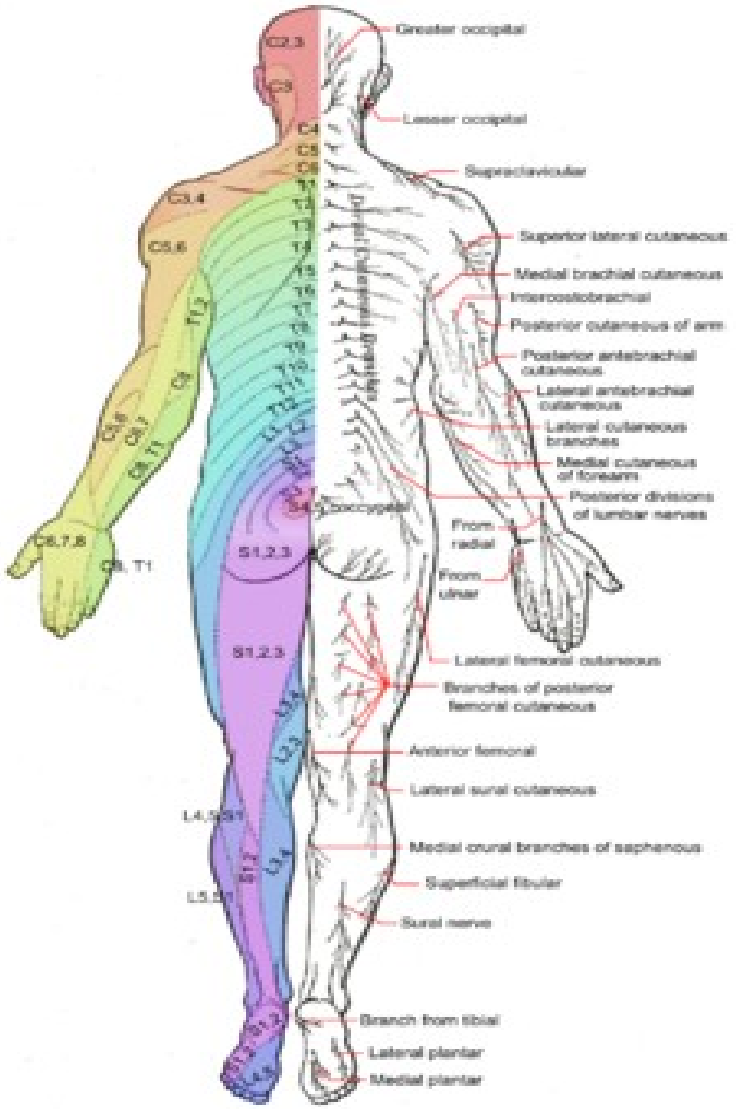
Nerve supply of the body wall

1. Supraclavicular nerves supply
    1. Above the 2nd rib & manubrosternal joint
2. Below this level midline & Para median strip is supplied by
    1. Anterior cutaneous branches of the spinal nerves from T2-L1



Nerve supply of the body wall

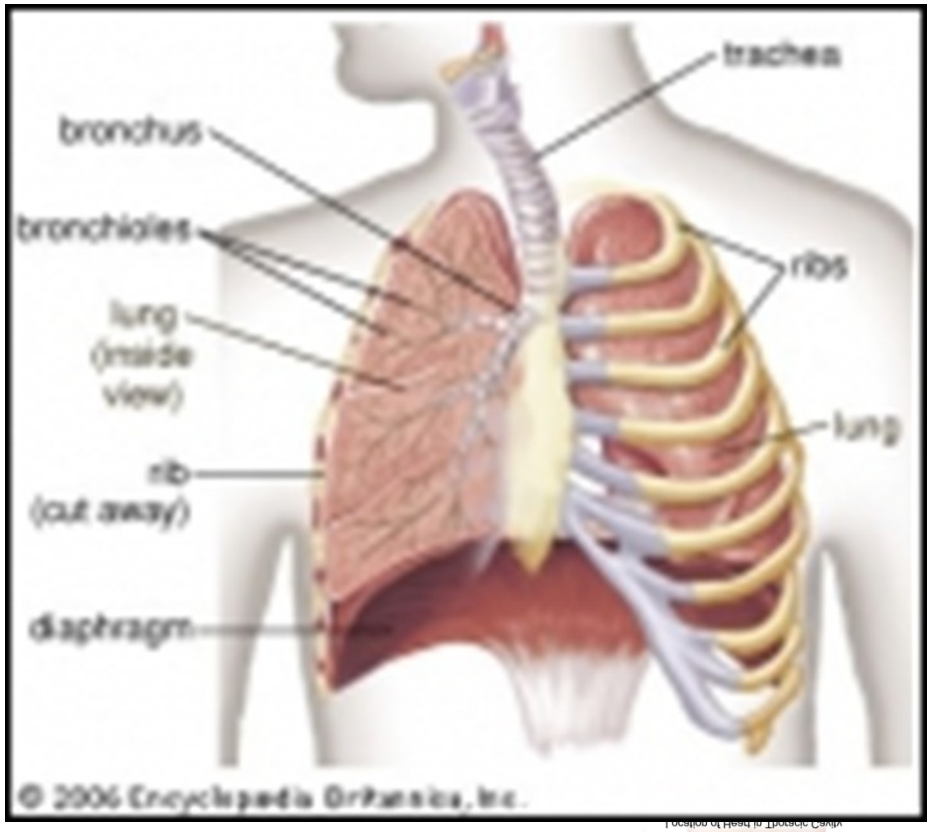
1. Skin in upper epigastric region is supplied by
    1. T7
2. Lateral strip by
    1. Lateral cutaneous branches of the spinal nerves T2-LI.
3. Posterior strip by
    1. Posterior rami of spinal nerves by their medial branches in upper thoracic & lateral branches in lower thoracic region.

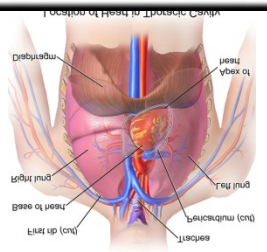


Thoracic wall & diaphragm

- Skeletal of the thoracic wall consists of

1. 12 thoracic vertebrae.
2. 12 pairs of the ribs & costal cartilages.
3. Sternum.
4. Thoracic cavity is roofed by suprapleural membrane.
5. Floored by diaphragm.

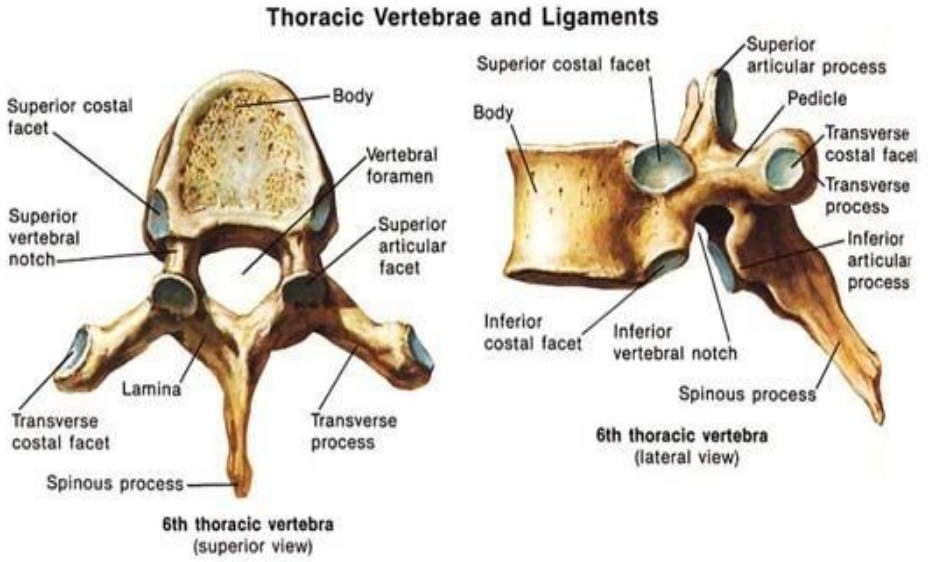




Thoracic vertebrae

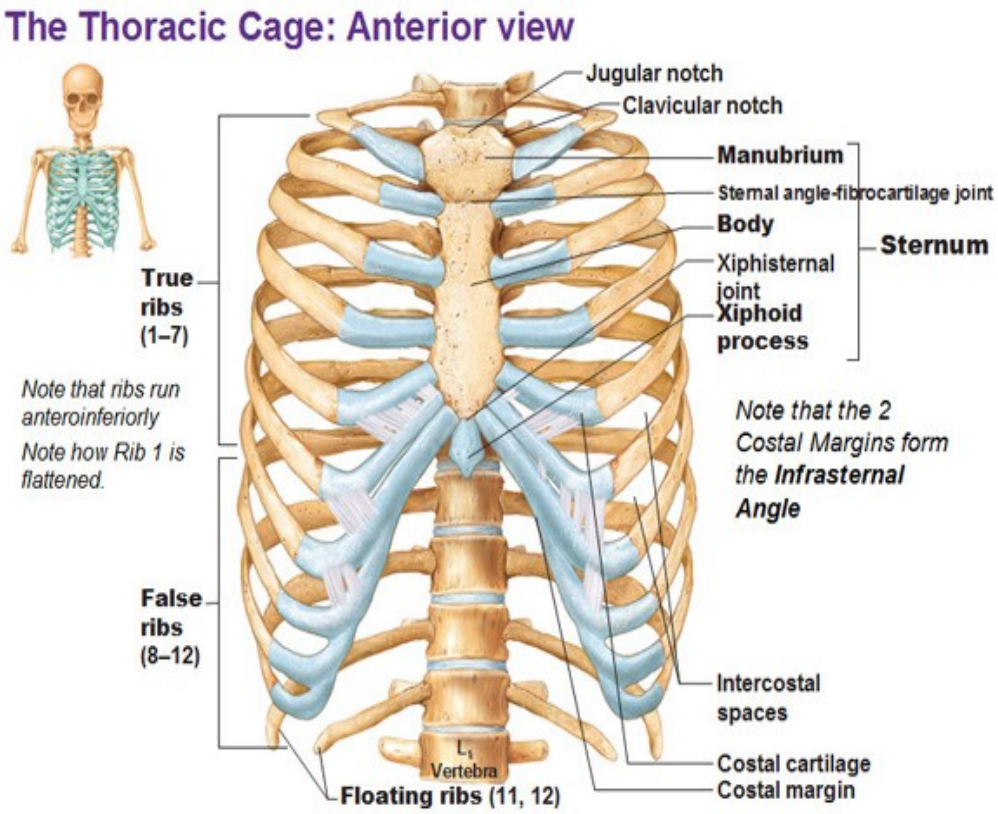
- 12 thoracic vertebrae.
- Identification by

1. Presence of costal facets on the sides of vertebral bodies .
2. 2-8 typical
3. 1st, 9th, 10th, 11th, 12th are atypical



ribs

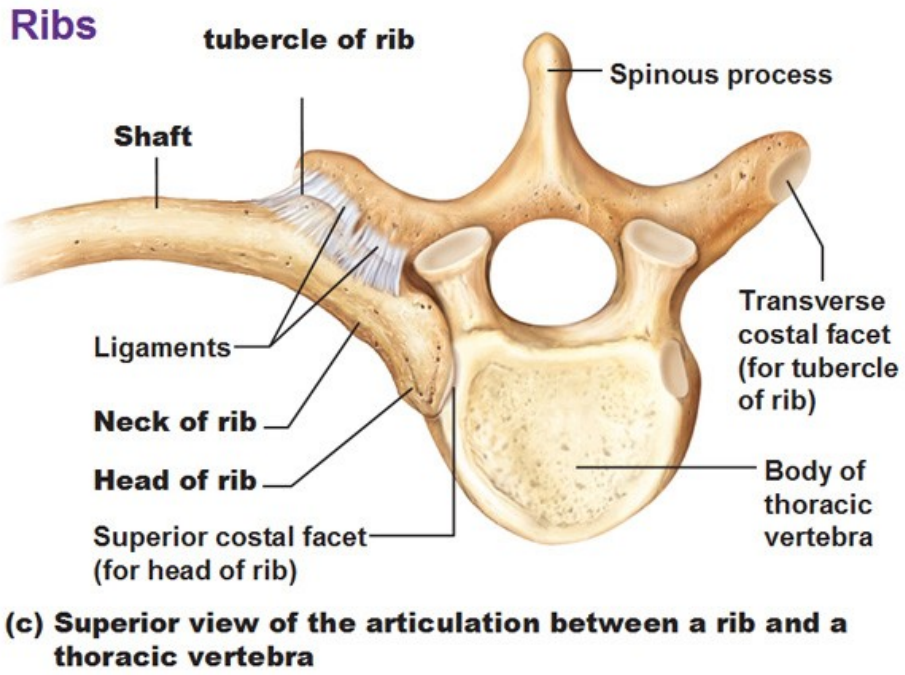
1. 12 pairs.
2. 1st seven true or vertebro sternal.
3. 8-12 false
    1. 8th, 9th & 10th vertebrochondral.
    2. 11th & 12th free floating
4. 3rd -9th typical
5. 1st, 2nd, 10th, 11th & 12th atypical.



Skeletal of thoracic wall

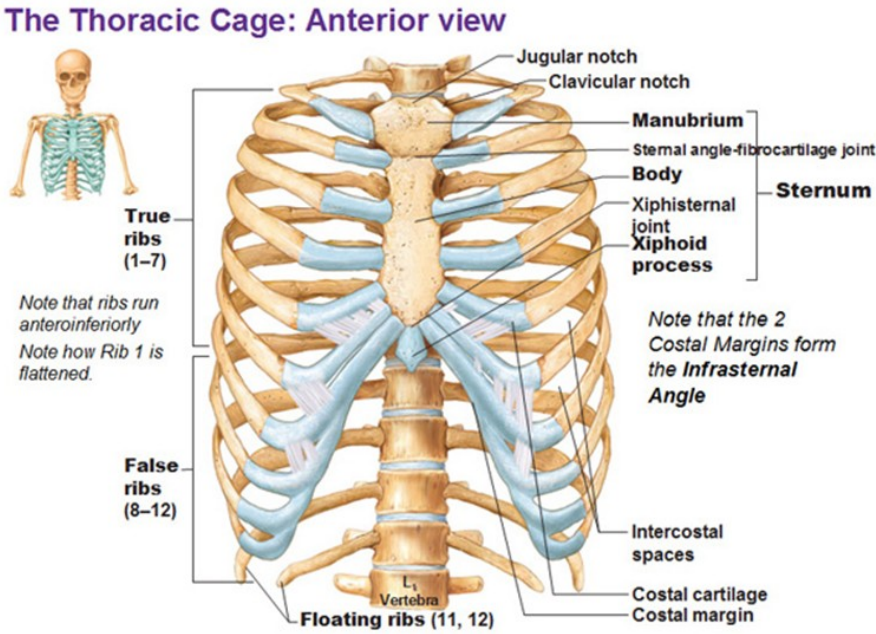
- Ribs articulates with the vertebral column in two places

1. By joints of costal heads & by costotransverse joints collectively these form the costovertebral joints



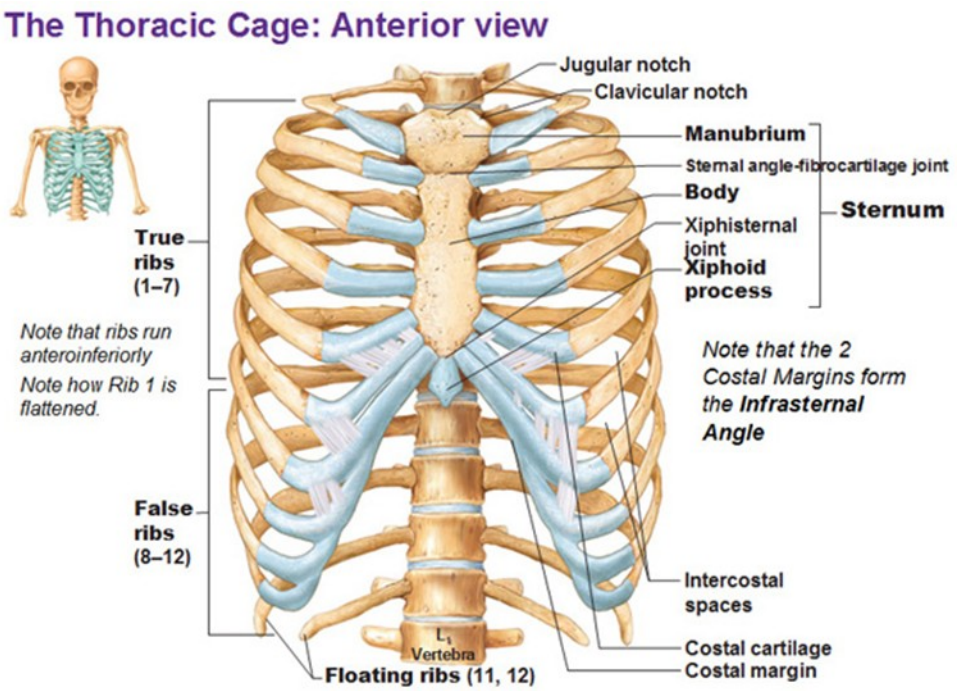
Skeletal of thoracic wall

1. At front ribs join their costal cartilages (costochondral) forms primary cartilaginous joint.
2. Upper seven costal cartilages articulate with sternum at the sternocostal joint
    1. 1st primary cartilaginous.
    2. 2nd-7th synovial



Skeletal of thoracic wall

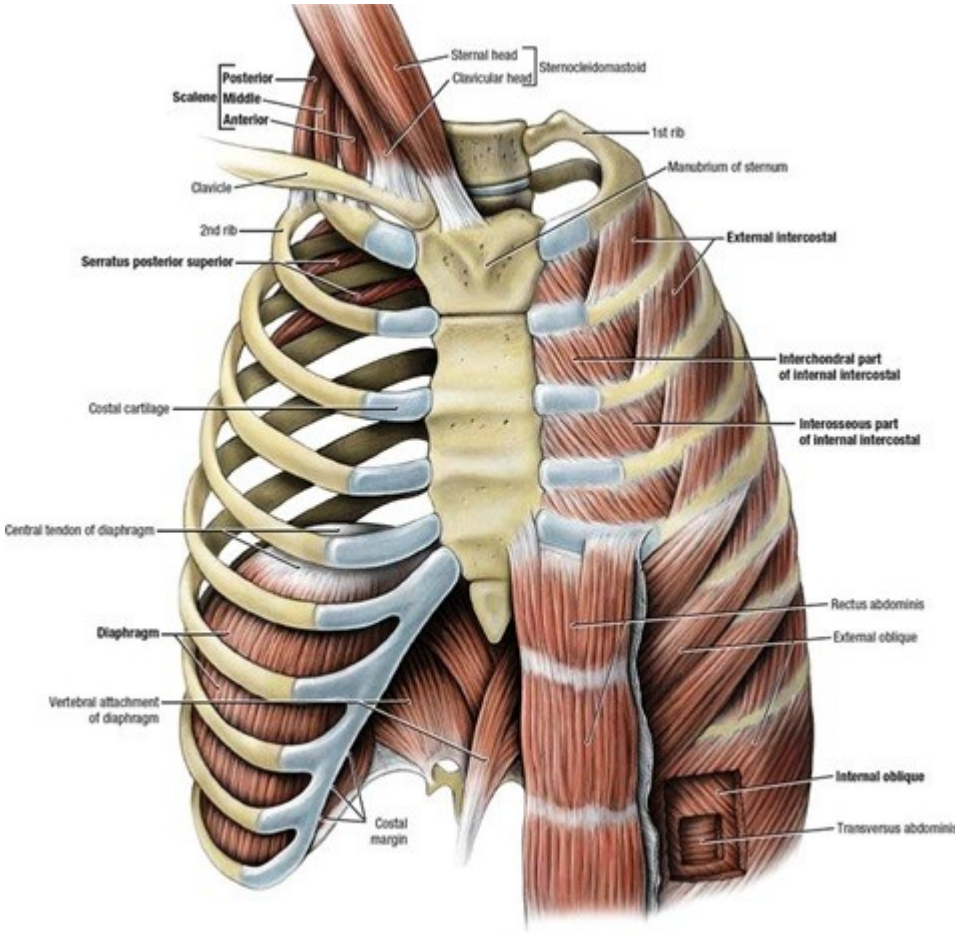
1. Adjacent surfaces of costal cartilages 6th & 7th , 7th & 8th , 8th &9th are joined at interchondral joints by synovial joint.
2. Last two costal cartilages are free.
3. Manubriosternal & xiphisternal joints are symphysis.



Thoracic muscles

- Muscles of the thoracic wall lie in three morphological layers.
- Innervated segmentally by anterior rami.

1. Outer layer .
    1. External intercostal
    2. Two small muscles have migrated posteriorly
        1. Serratus posterior superior & inferior.



Intercostal muscles , vessels & nerves

objectives

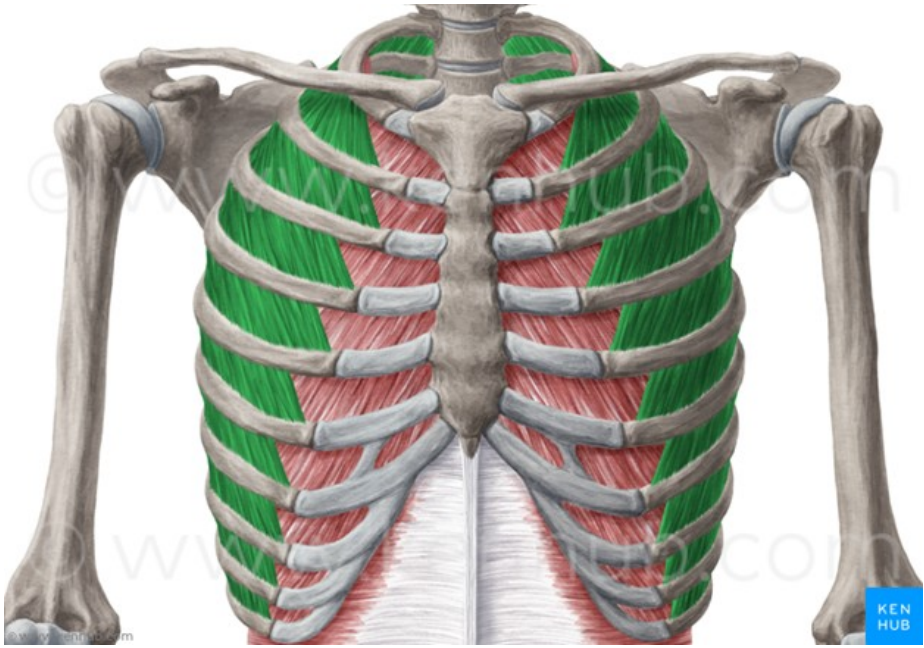
- At the end of class students should be able to understand.

1. Origin , insertion, action & nerve supply of intercostal muscles.
2. Intercostal arteries
3. Intercostal veins.

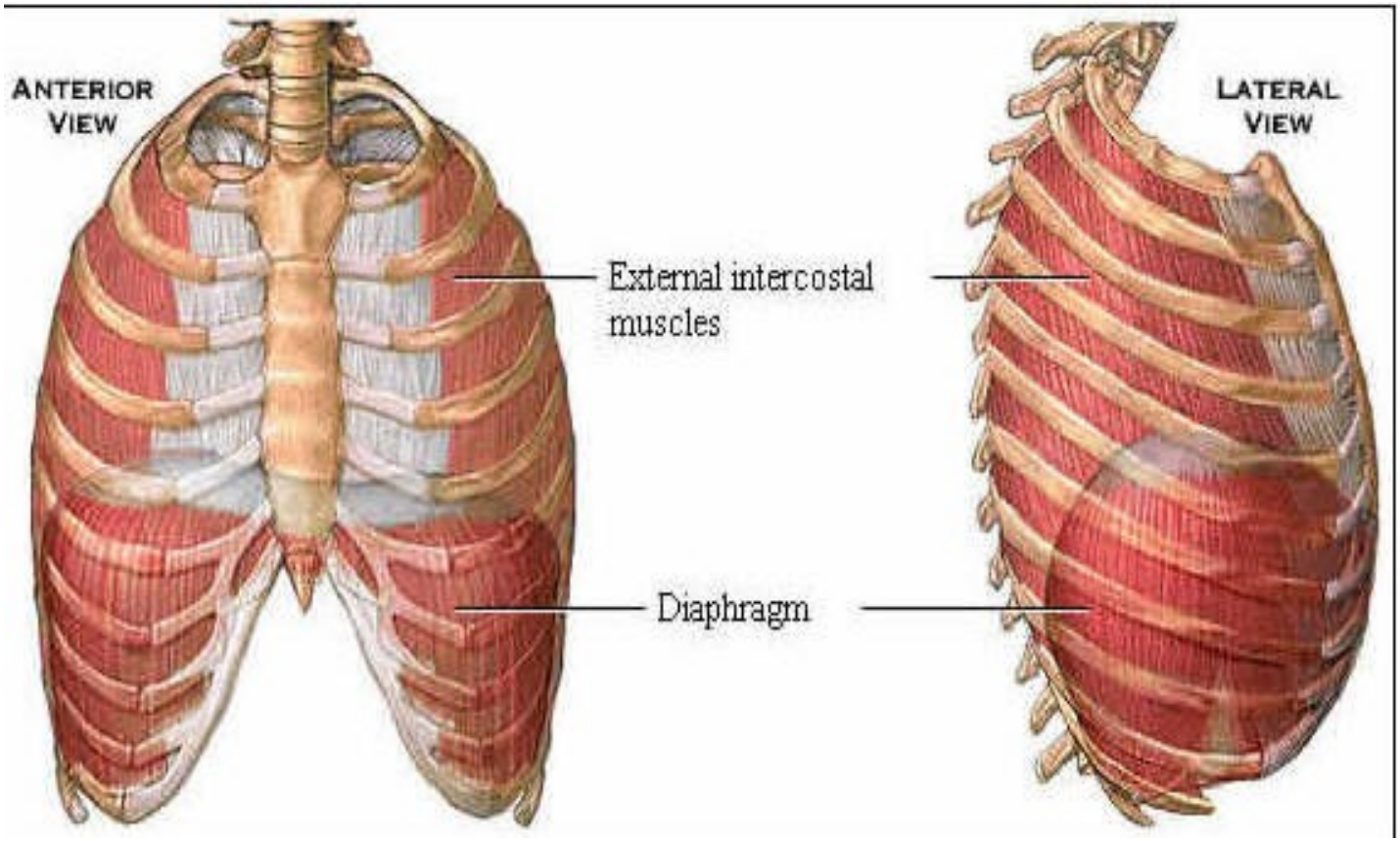
Chief muscles of the outer layer

- External intercostal muscles.

1. Fibres pass obliquely downwards & forwards from sharp lower border of rib above to the smooth upper border of the rib below.
2. Each muscle extends from superior costotransverse ligament at back of intercostal space to costochondral junction.
3. Replaced by external intercostal membrane.
4. That extends to the side of the sternum.



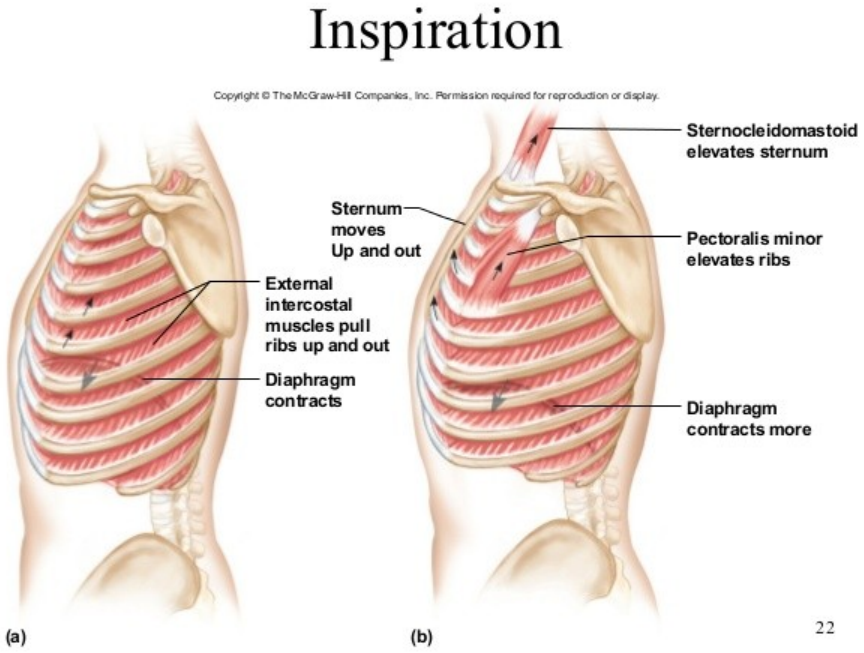
External intercostal muscles



Middle layer

- Internal intercostal muscles.

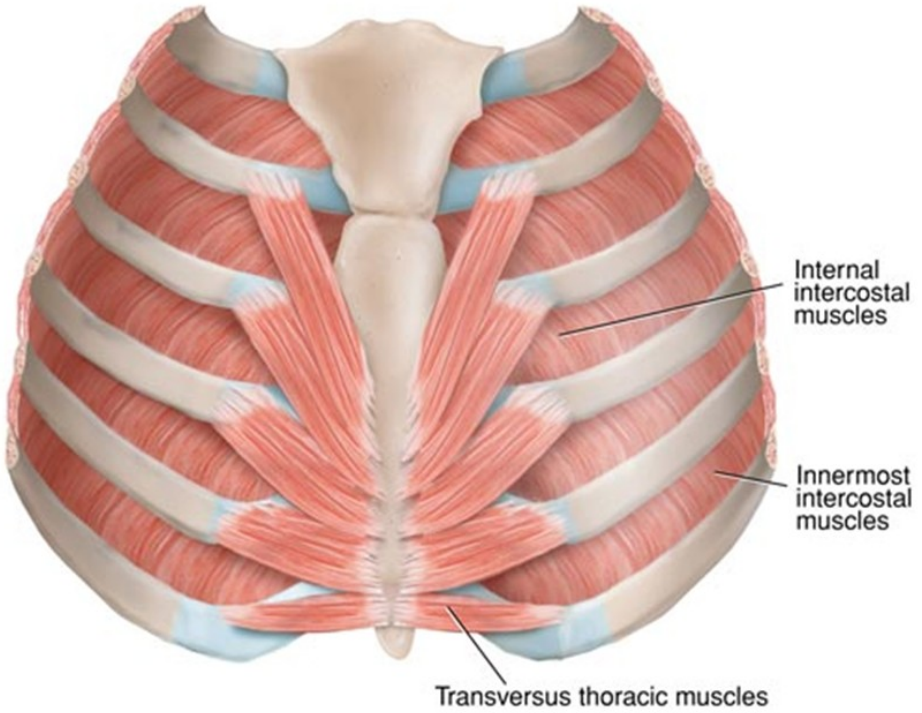
1. Fibres run downwards & backwards.
2. From costal groove to the upper border of rib below.
3. Each muscle extends anteriorly to the side of the sternum & replaced posteriorly by posterior intercostal membrane.
4. Membrane extends from the angle of the rib to the superior costotransverse ligament.



Inner layer (transversus thoracis)

- Three groups of the muscles in this layer.

1. Inner most intercostals
    1. Lines the rib cage at the side.
2. Subcostalis
    1. At the back.
3. Sternocostalis.
    1. At front.

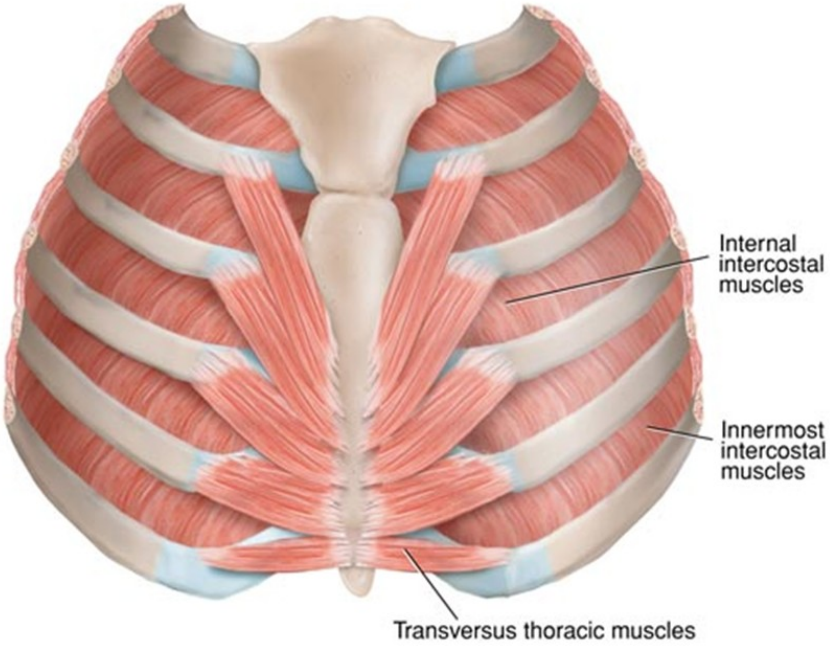


View from within thoracic cavity

Transversus thoracic

- sternocostalis.

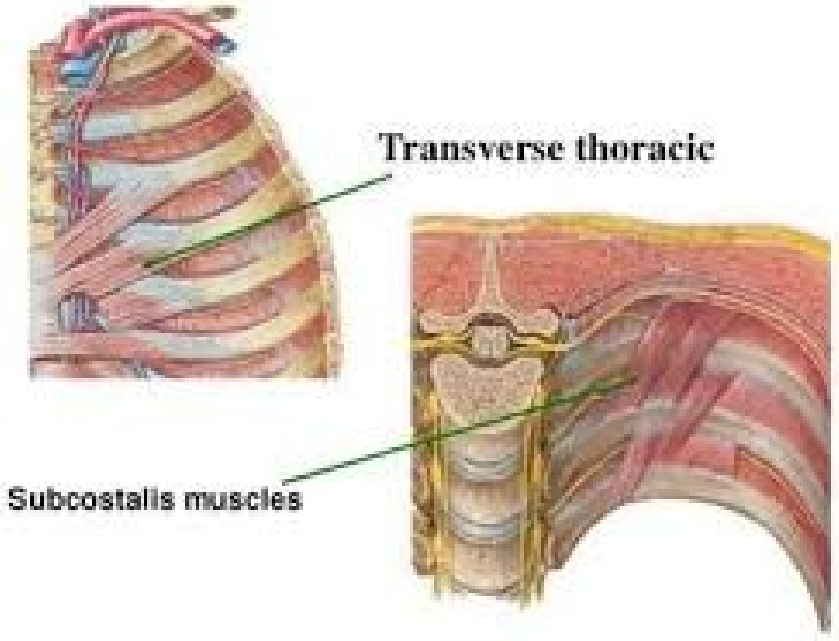
1. Origin from
    1. Posterior surface of the sternum.
    2. Digitations diverge on the each side to the 2nd to 6th costal cartilages.



View from within thoracic cavity

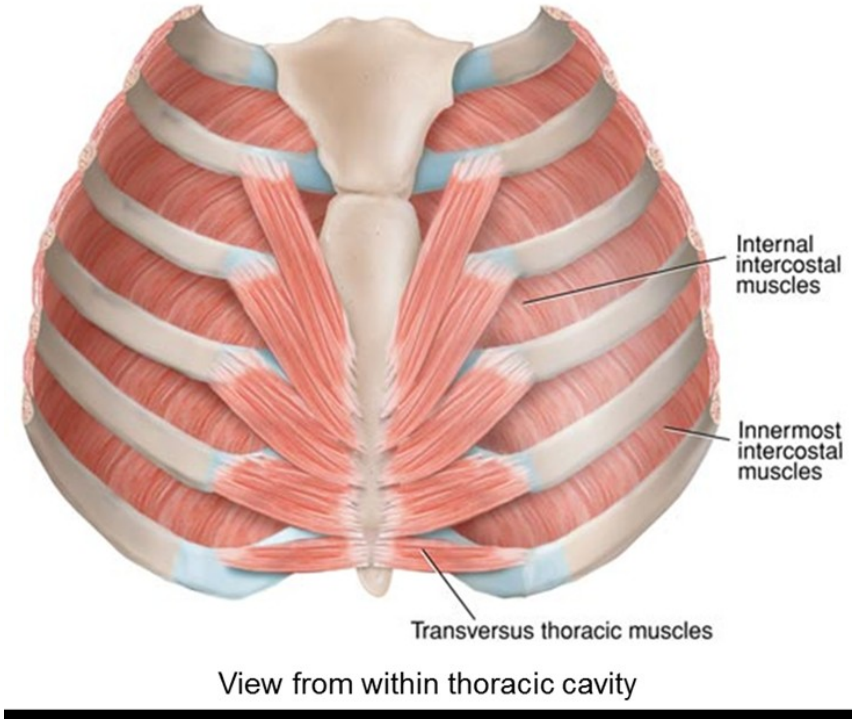
Transversus thoracic

- Subcostalis.
- 0rigin
  - Inner surface of the rib near the angle.
- Insertion.
  - Inner surface of 2-3 ribs below.



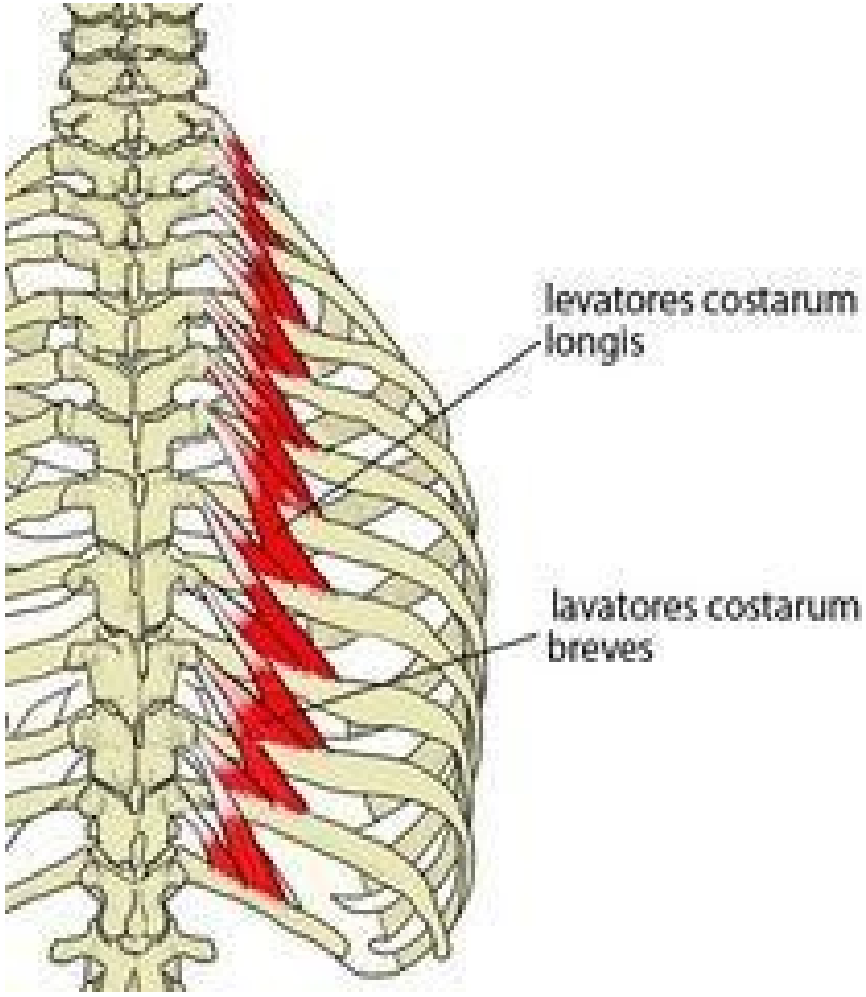
Transversus thoracic

- Intercostalis intimi
- Origin.
  - Middle 2/4 of ridge above the costal groove.
- Insertion.
  - Inner lip of the upper border of the rib below.



Levator costae

- Each one of the 12 pairs is fan shaped.
- Origin
  - Tip of the transverse process from C7 – T11. vertebra.
- Insertion.
  - Upper border of the rib below lateral to the tubercle
- Nerve supply
  - Posterior rami of C8- T11.

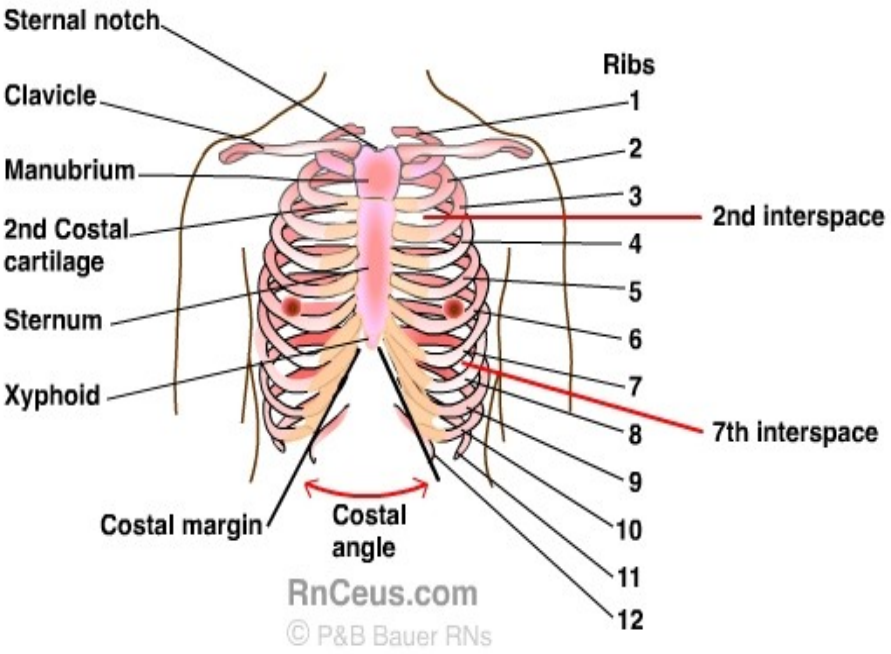


Action of intercostal muscles

1. Main action of the intercostal muscles is to prevent retraction of intercostal spaces during expiration & their bulging outwards during inspiration.
2. External intercostal ,interchondral portion of the internal intercostals & levator costae elevate the ribs during inspiration.
3. Internal intercostal except the interchondral portion & transversus thoracis depress the ribs or cartilages during expiration.

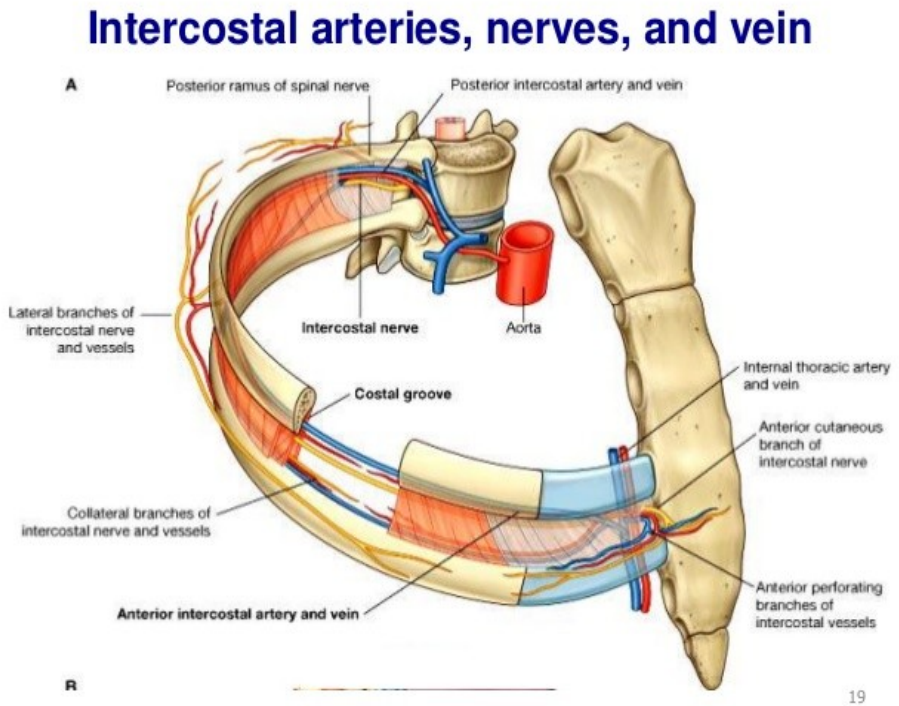
Intercostal space

1. Space b/w ribs.
2. Filled in by three layers of muscles.
3. Vein , artery & nerve lie b/w intermediate & inner layers.
4. Needle for pleural drainage is inserted just above the rib that forms lower boundary of space.
5. Collateral branches of nerve & vessels run along the upper border.



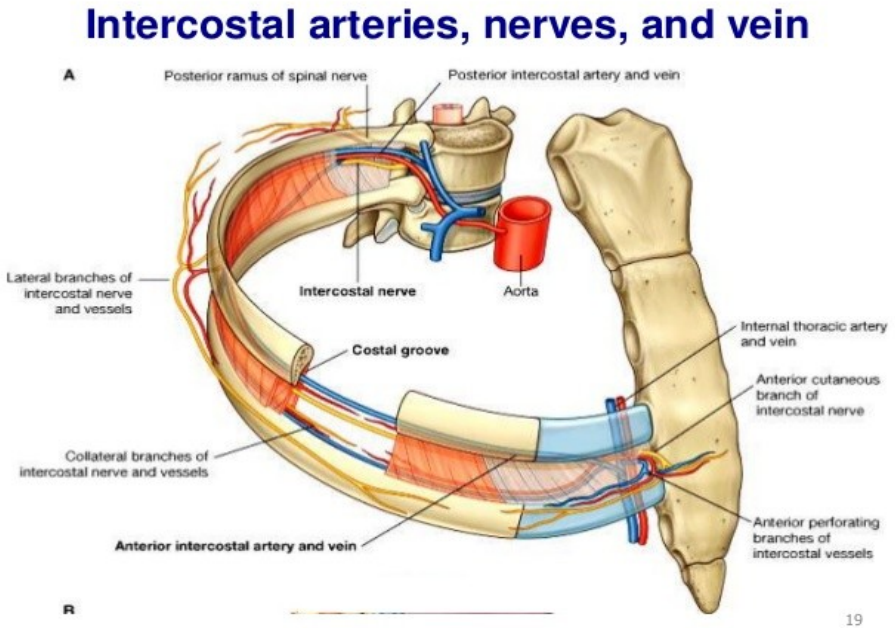
Intercostal nerves

1. Mixed spinal nerve.
2. Emerged from the intervertebral foramen.
3. Give posterior ramus.
4. Pass around neurovascular plane b/w internal intercostal & transversus thoracic



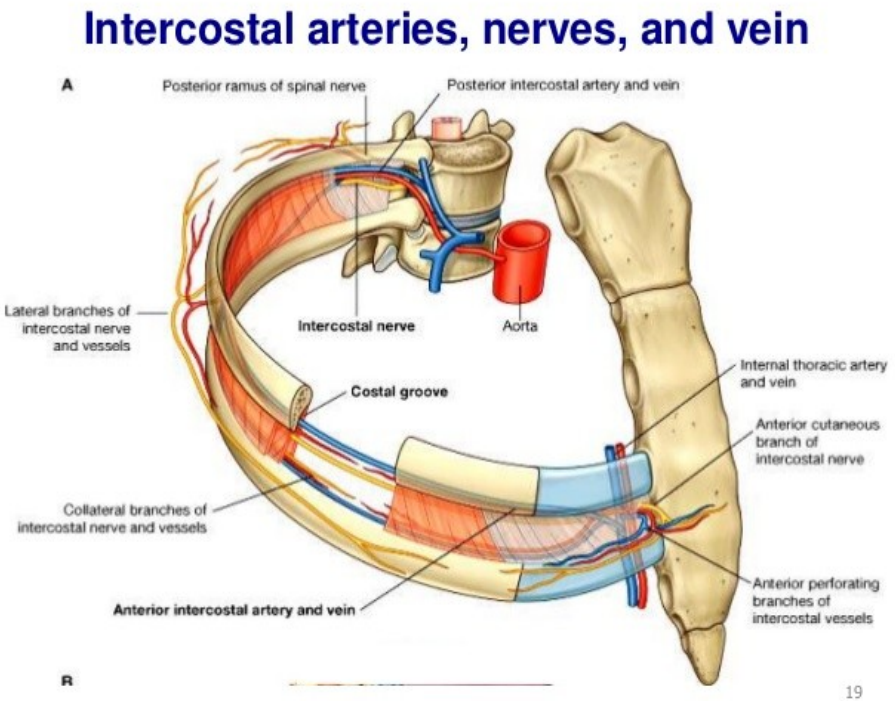
Intercostal nerve

- Gives of
  - collateral & lateral cutaneous branch before costal angle.
- Collateral branch runs along the inferior border of the space supplies the muscles of the space , parietal pleura & periosteum of the ribs.



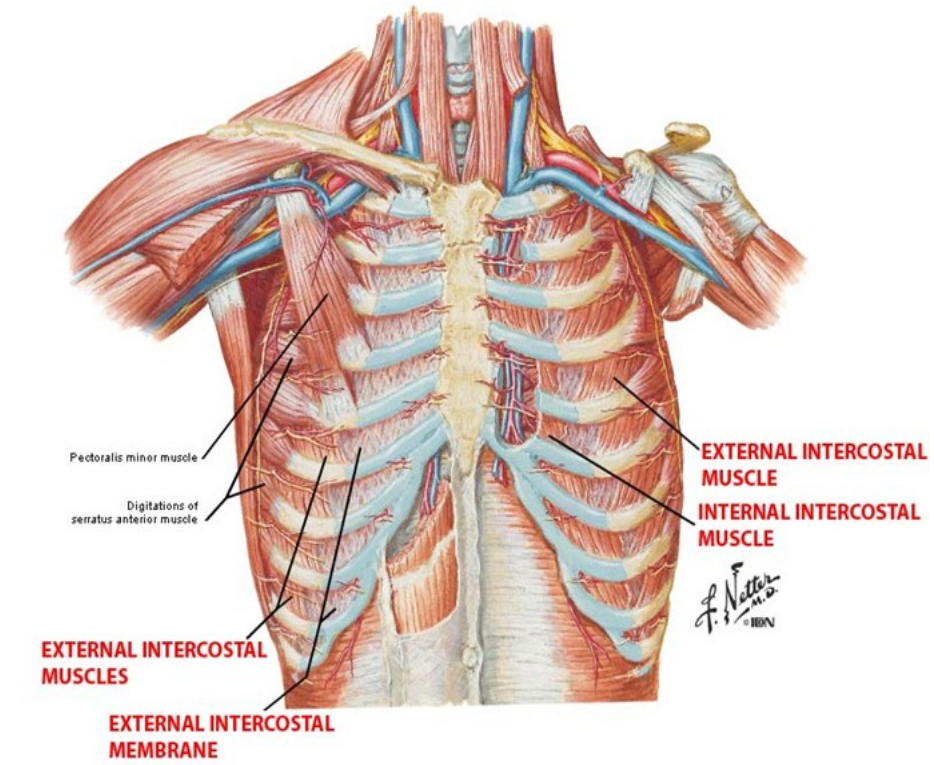
Intercostal nerve

- Lateral cutaneous branch pierces the intercostal muscles & overlying muscles of the body wall along the mid axillary line divides into
  - Anterior & posterior branch to supply the skin over the space



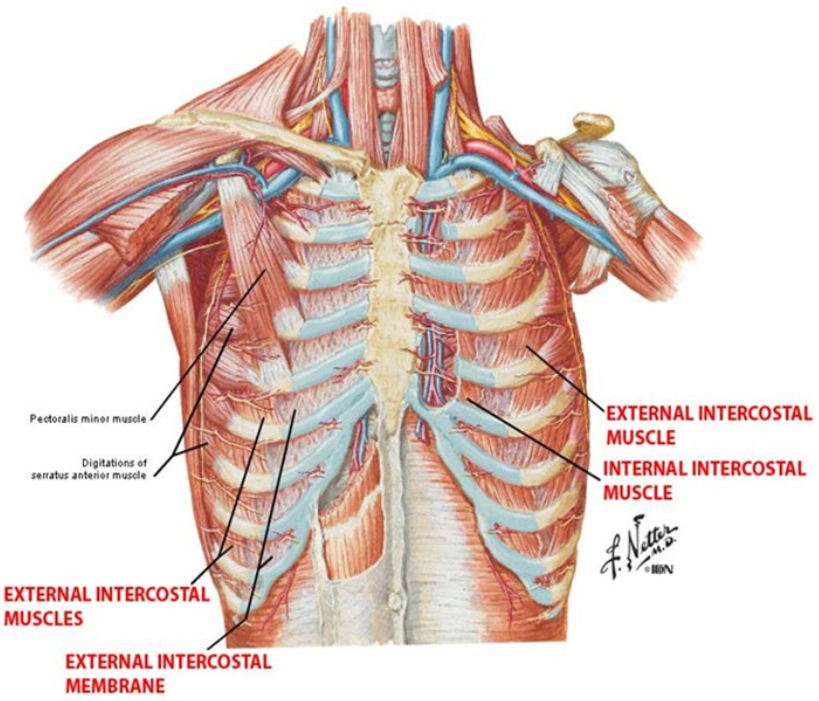
Intercostal nerve

- Ends as an anterior cutaneous nerve , which passes anterior to internal thoracic artery in upper six spaces.
- Pierce the intercostal muscles to supply skin.



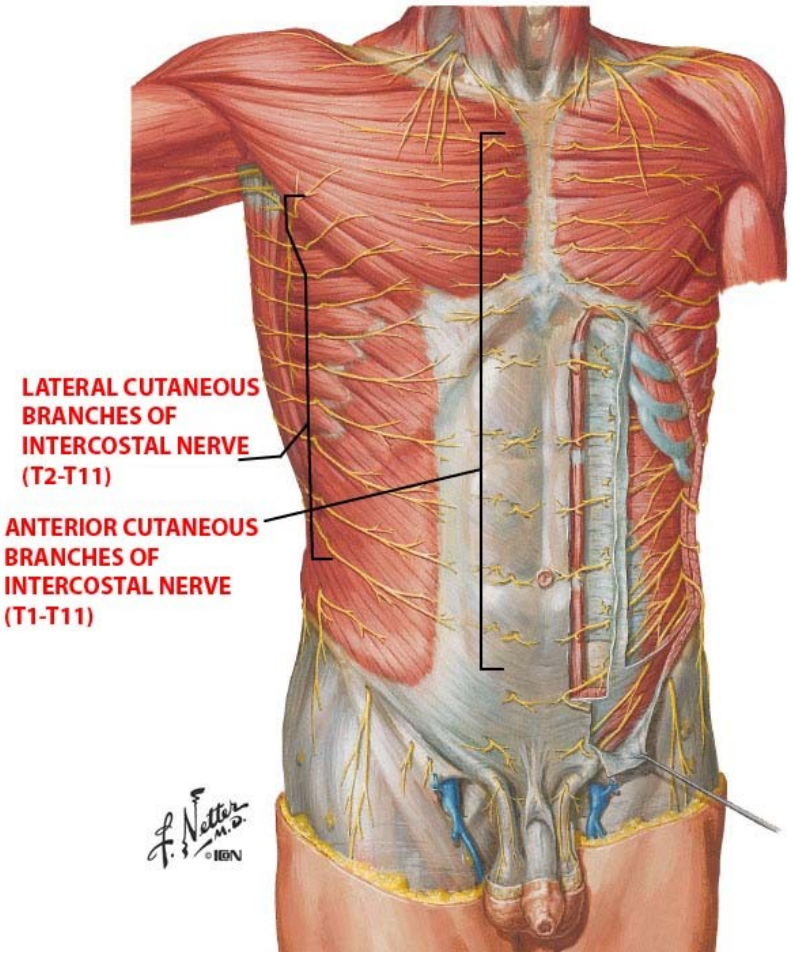
Intercostal nerves

- Anterior ramus of T1 ascends across the neck of the 1st rib to enter into brachial plexus.
- Before it gives small 1st intercostal nerve which course around beneath the inferior surface of the 1st rib to supply intercostal muscles, pleura & periosteum of 1st space.



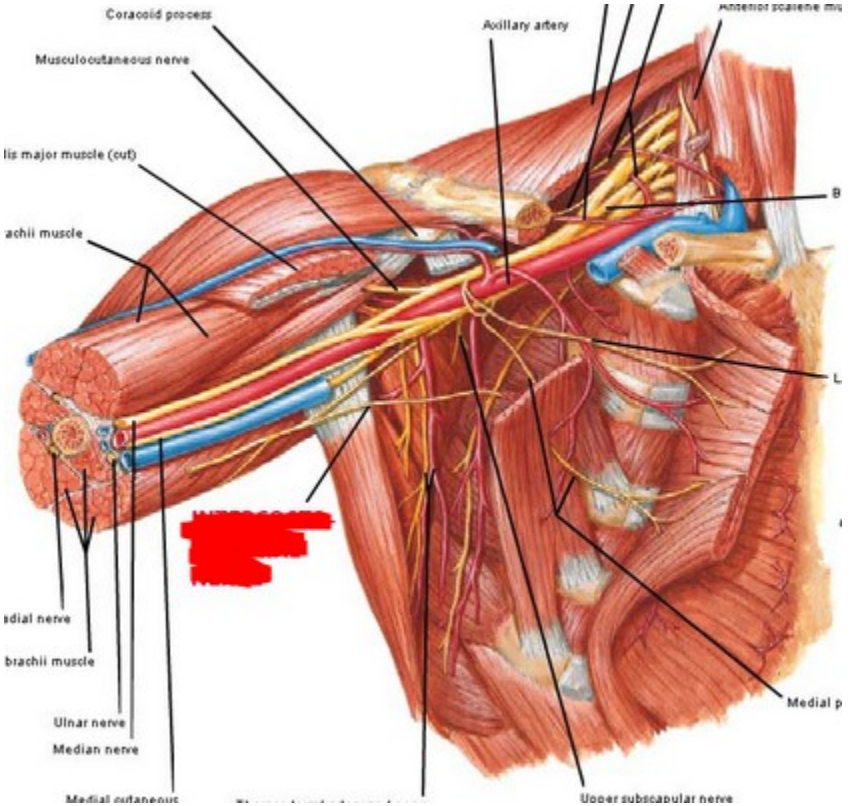
Intercostal nerves

- Lower five intercostal nerves & their collateral branches slopes downwards behind the costal margins into neurovascular plane of the abdominal wall to which the supply.



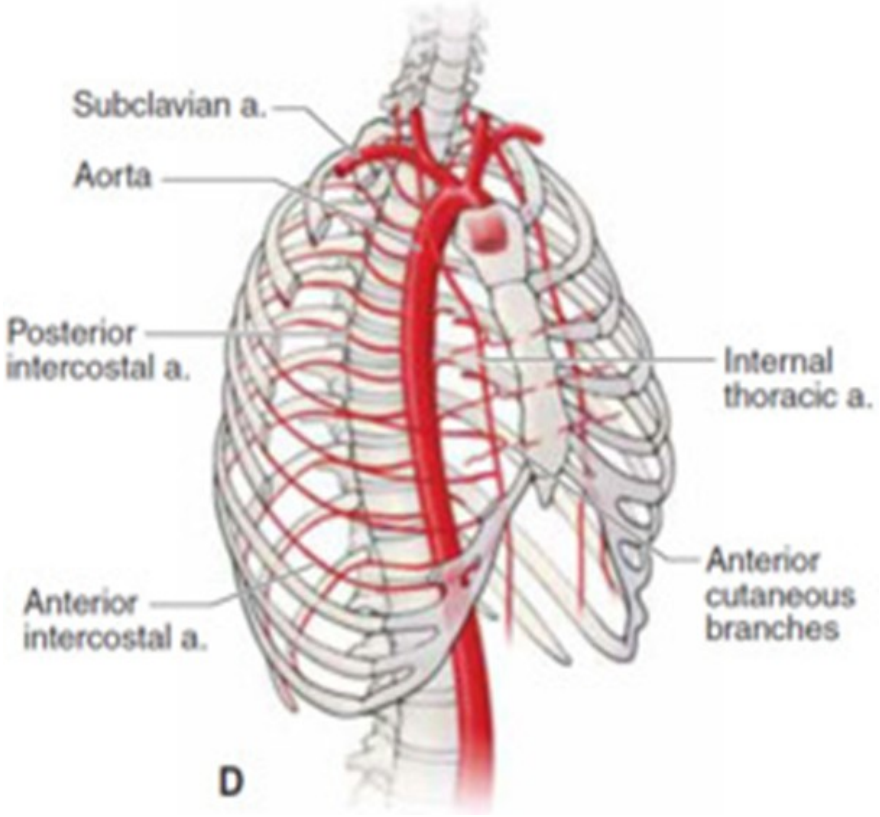
Intercostal nerves

- Lateral cutaneous branch of 2nd intercostal nerve crosses the axilla to medial side of the arm as intercostobrachial nerve
- Subcostal nerve passes behind the lateral arcuate ligament.



INTERCOSTAL ARTERIES

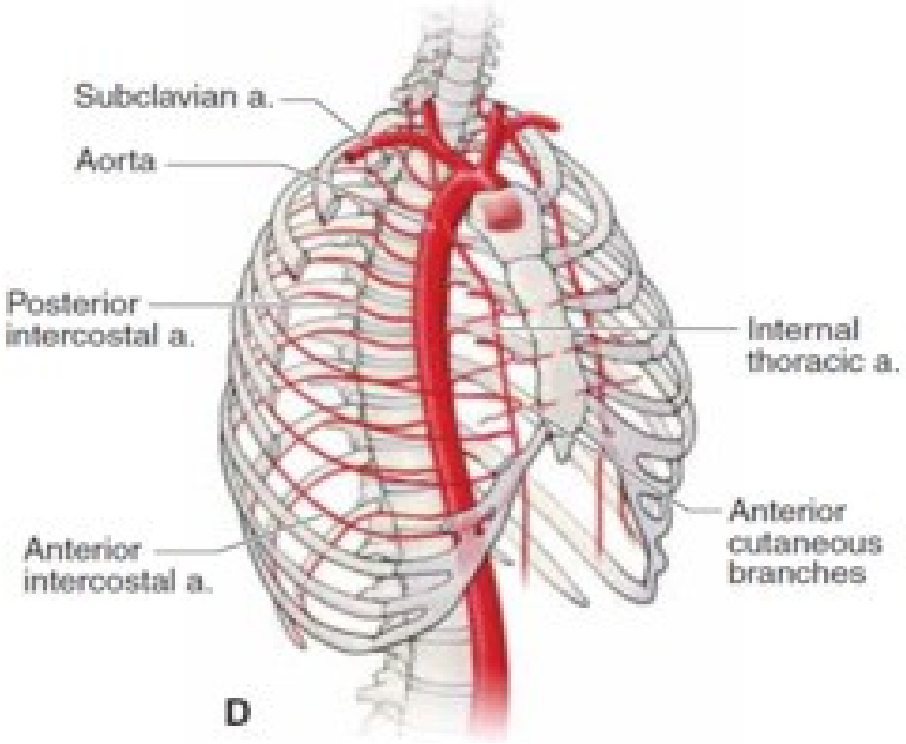
- Each intercostal space contains one posterior intercostal artery & two anterior intercostal arteries.
- Greater part of space is supplied by posterior intercostal arteries.



Intercostal arteries

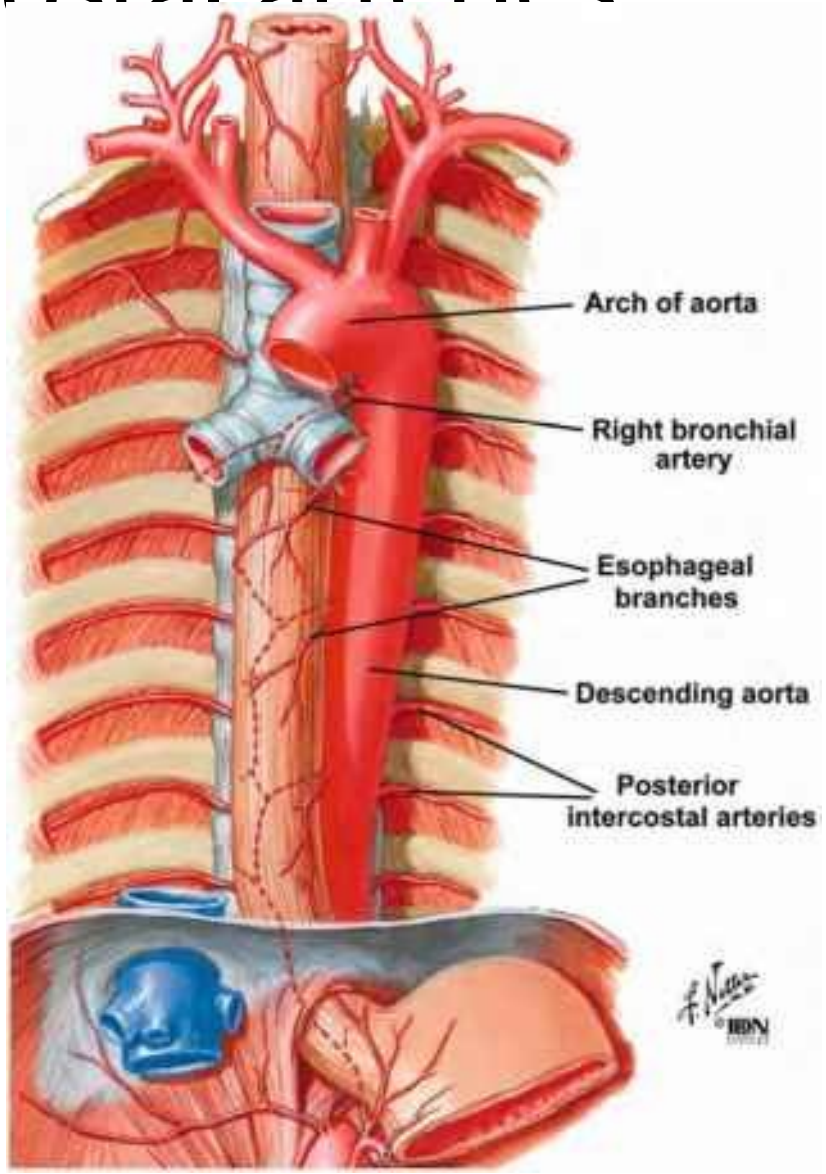
- Anterior intercostal arteries.

1. Upper six from internal thorax artery.
2. 7th, 8th & 9th spaces from musculophrenic.
3. No intercostal arteries in lower two spaces.



Posterior intercostal arteries

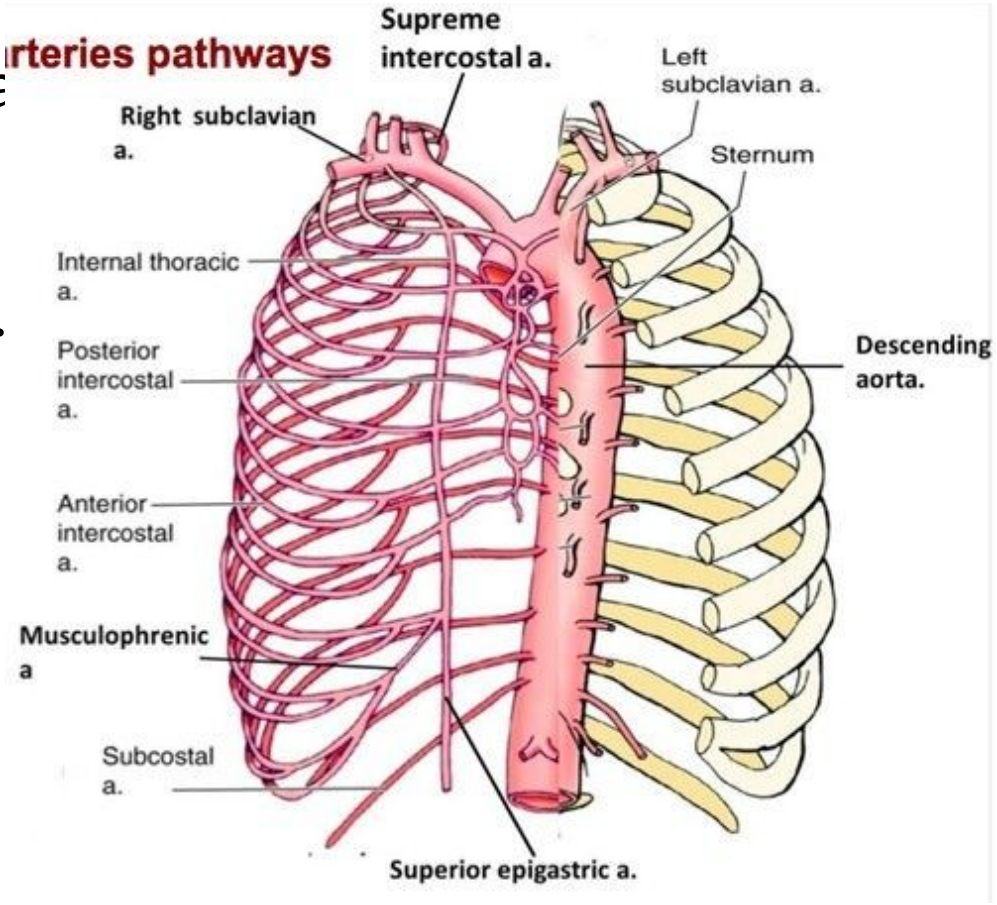
1. 11 in number.
2. Upper two intercostal spaces are supplied by superior intercostal artery.
3. Lower nine intercostal spaces are supplied by descending thoracic aorta.



Course of posterior intercostal arteries

- IN front of the vertebrae.

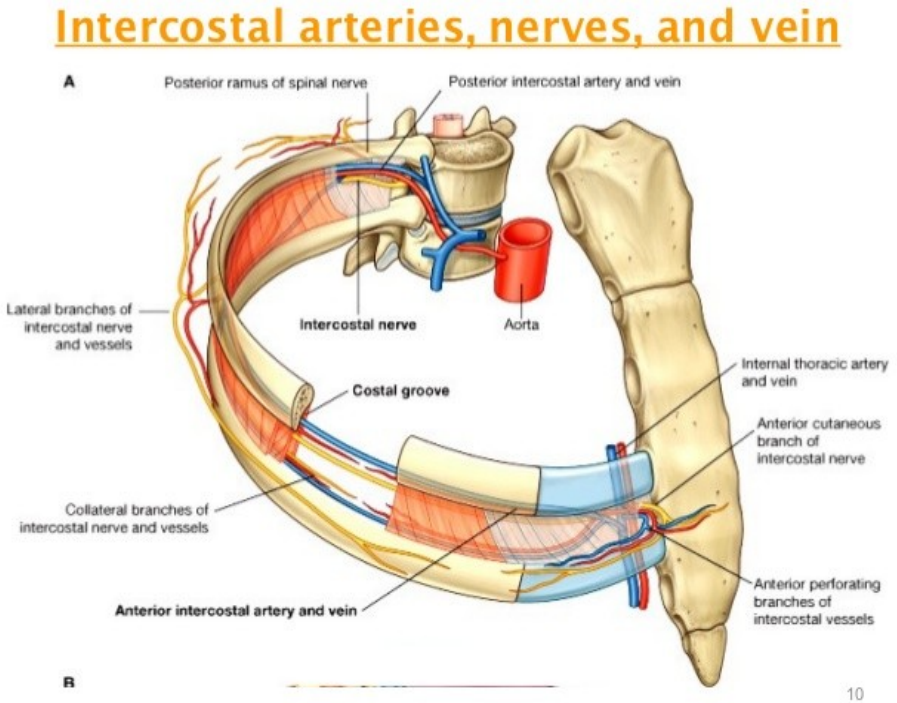
1. Right posterior intercosta arteries are larger than left.
2. Pass behind oesophagus. Thoracic duct, azygous vein & sympathetic chain .
3. Left posterior arteries pass behind the hemizygous vein & sympathetic chain.



Course of posterior intercostal arteries

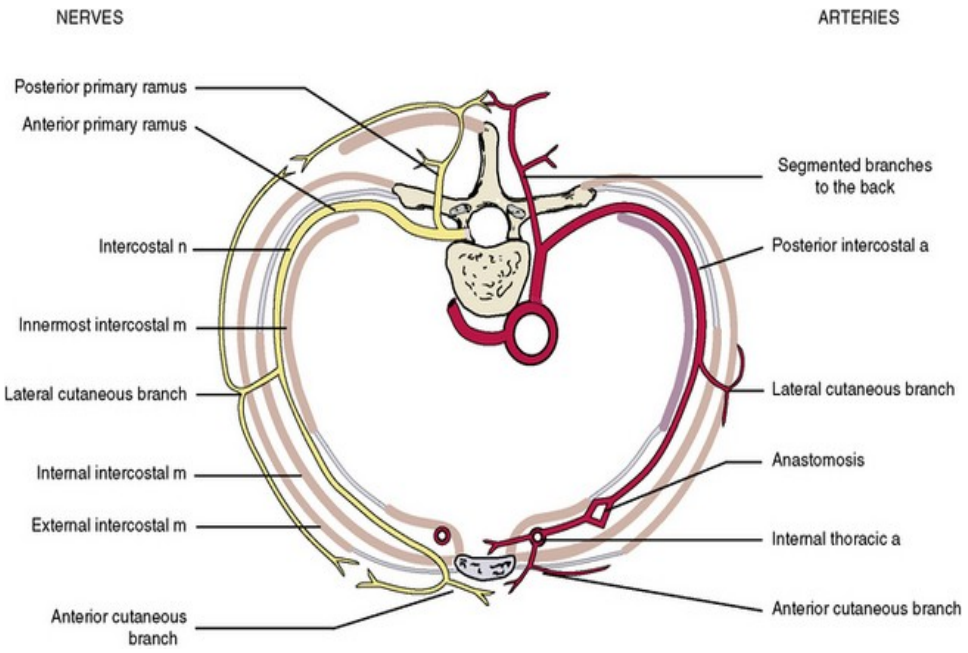
- In intercostal space.

1. Accompanied by vein & nerve (VAN)
2. Run in costal groove 1st B/W pleura & internal intercostal membrane & than B/W internal intercostal & intercostalis intimi muscles.
3. Terminate at costochondral junction by anastomosing upper anterior intercostal artery.



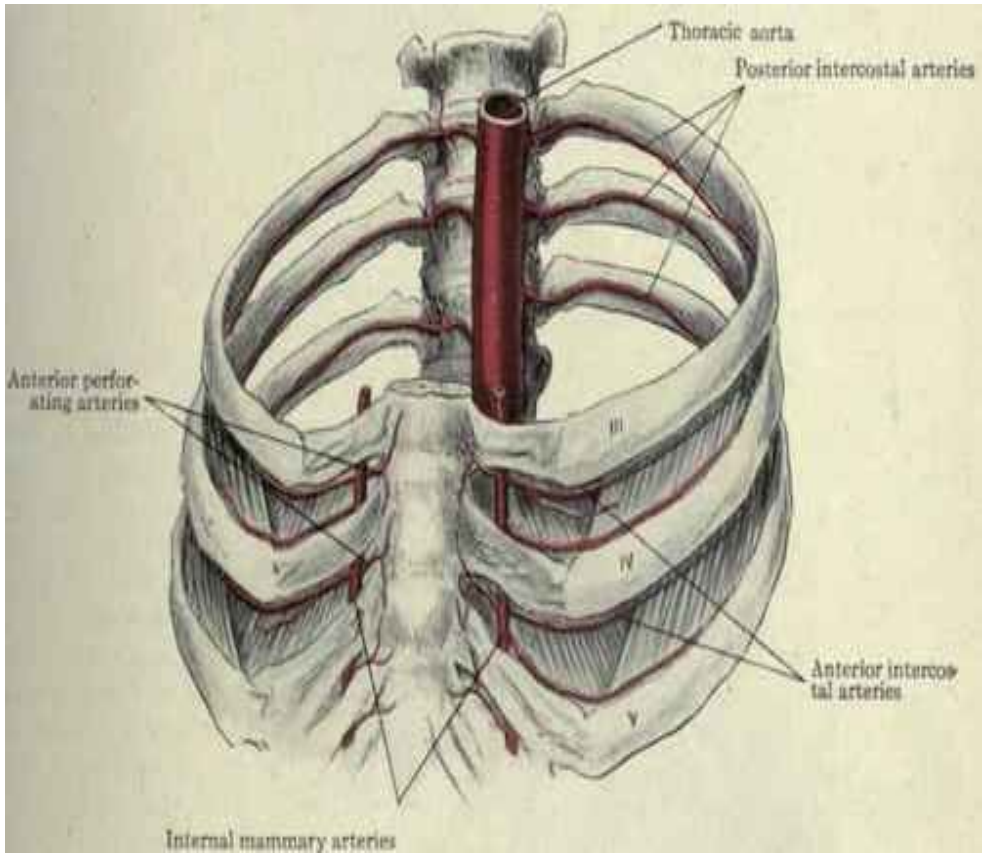
Branches of posterior intercostal arteries

1. Dorsal branch supplies
    1. Skin & muscles of the back.
    2. Spinal branch to spinal cord & vertebrae.
2. Collateral branch
    1. Anastomose with lower anterior intercostal artery
3. Muscular branches
    1. Intercostal , pectoral & serratus anterior.



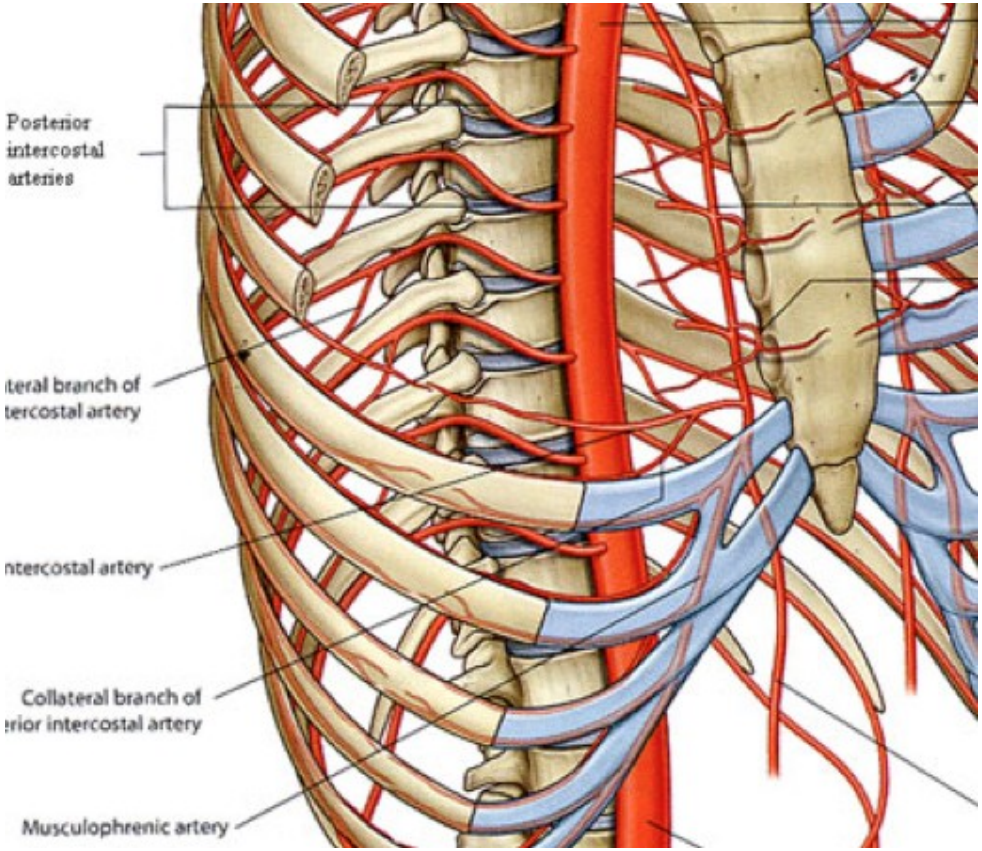
Branches of posterior intercostal arteries

- 4. lateral cutaneous.
- 5. mammary branches from 2nd to 4th supply breast
- 6. right bronchial artery arise from right 3rd posterior intercostal artery.



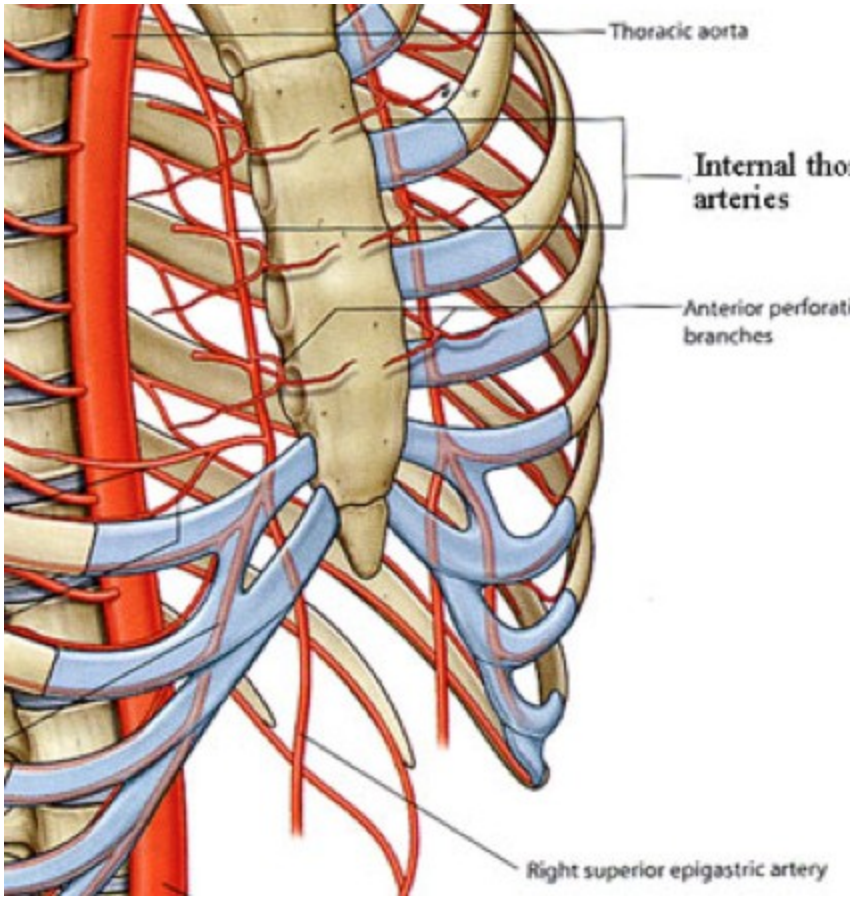
Internal thoracic artery

1. Arise from 1st part of subclavian artery.
2. 2cm above the sternal end of the clavicle.
3. Terminate in 6th intercostal space by dividing into
    1. Superior epigastric
    2. Musculophrenic .



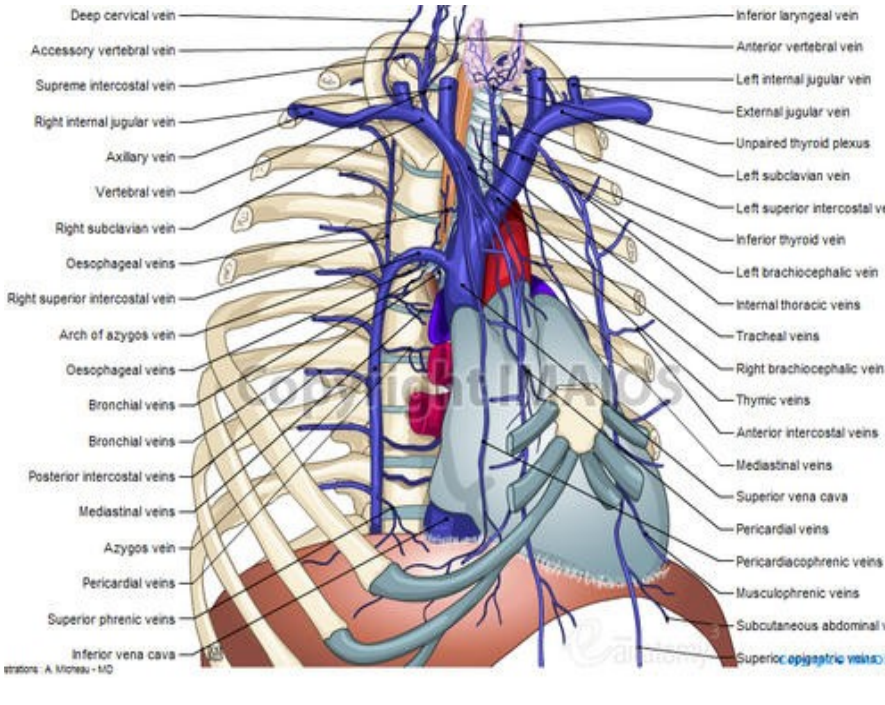
Branches of internal thoracic arteries

1. Pericardiophrenic
    1. Pleura & pericardium
2. Mediastinal arteries
3. Two anterior intercostal arteries in each of upper six spaces.
4. Perforating in 2nd to 4th space to breast.



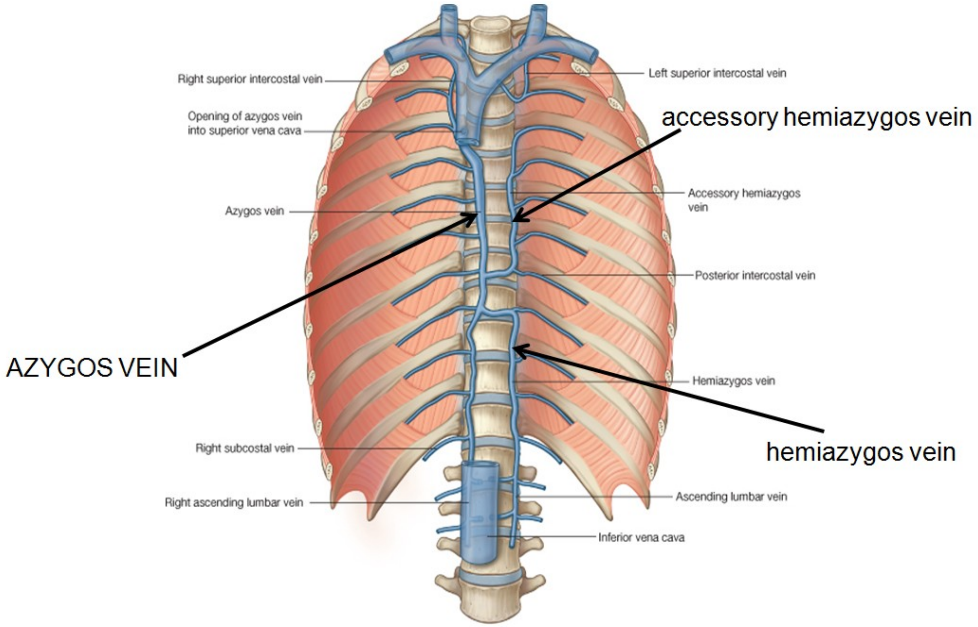
intercostal veins

- In each intercostal space are one posterior & two anterior intercostal veins.
- Anterior veins drains into internal thoracic & musculophrenic.



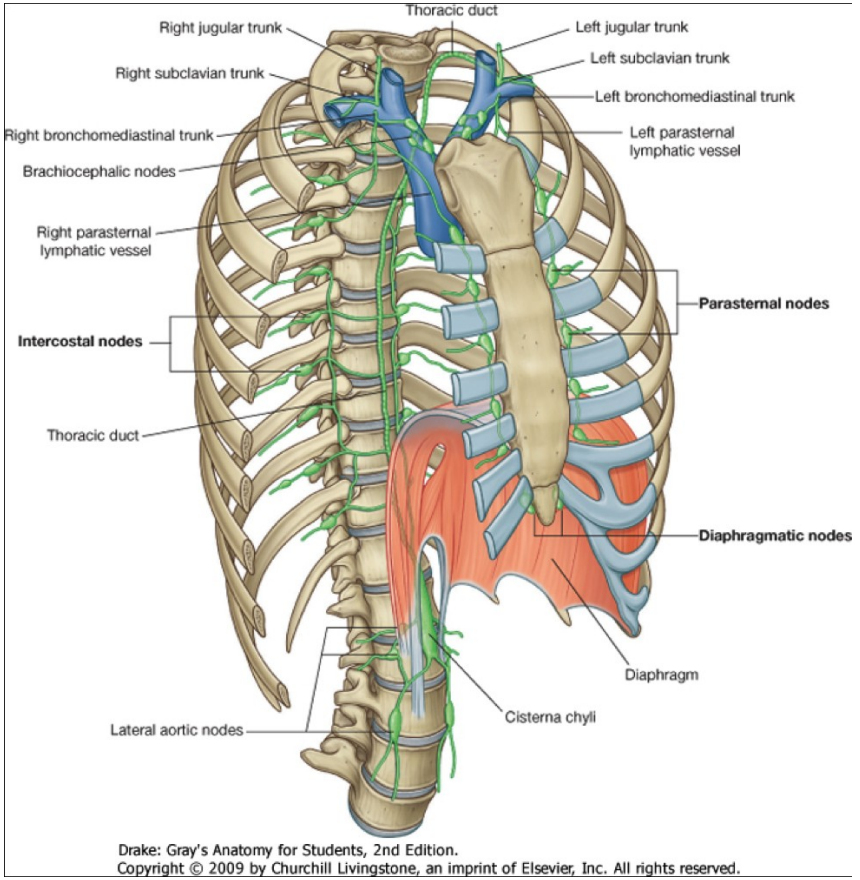
Posterior intercostal veins

1. 1st posterior vein drains into vertebral or brachiocephalic.
2. 2nd , 3rd & sometime 4th form superior intercostal vein.
    1. On right drains into azygos vein.
    2. On left drains into left brachiocephalic vein.
3. Lower 8 spaces
    1. On right side drains into azygos vein
    2. On left drains into hemiazygos & accessory hemiazygos.



Lymphatic drainage

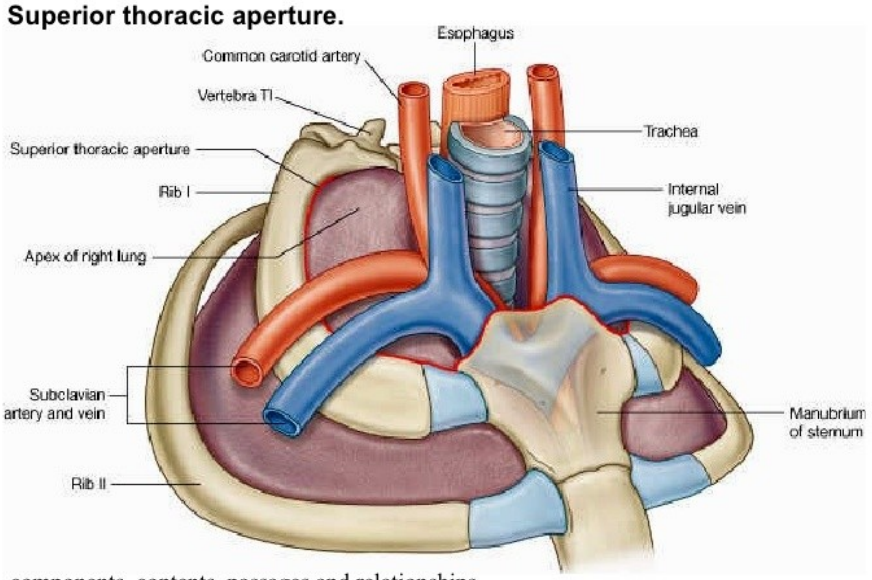
- Anteriorly intercostal pace drains into anterior intercostal nodes (parasternal).
- Posteriorly into posterior intercostal nodes.



THANK YOU

Thoracic inlet

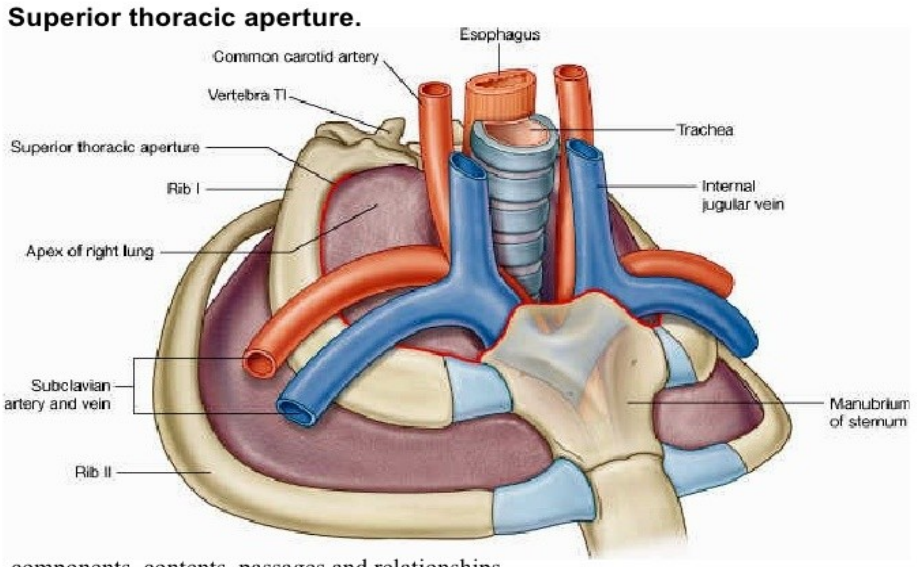
1. Narrow upper end of the thorax that is continuous with the neck.
2. Is kidney shape.
3. Transverse diameter is 10-12.5 cm.
4. A.P diameter is 5cm



components, contents, passages and relationships
Superior thoracic aperture: vertebra TI, rib I, costal cartilage I, manubrium of sternum
Contents: Apices of right and left lung; brachiocephalic, subclavian arteries and veins; trachea, esaphagus; vagus nerves; sympathetic trunc; phrenic nerves

Boundaries of thoracic inlet

- Anteriorly.
  - Upper border of manubrium sterni.
- Posteriorly.
  - Superior surface of the body of 1st thoracic vertebra.
- On each side
  - 1st rib & cartilage.

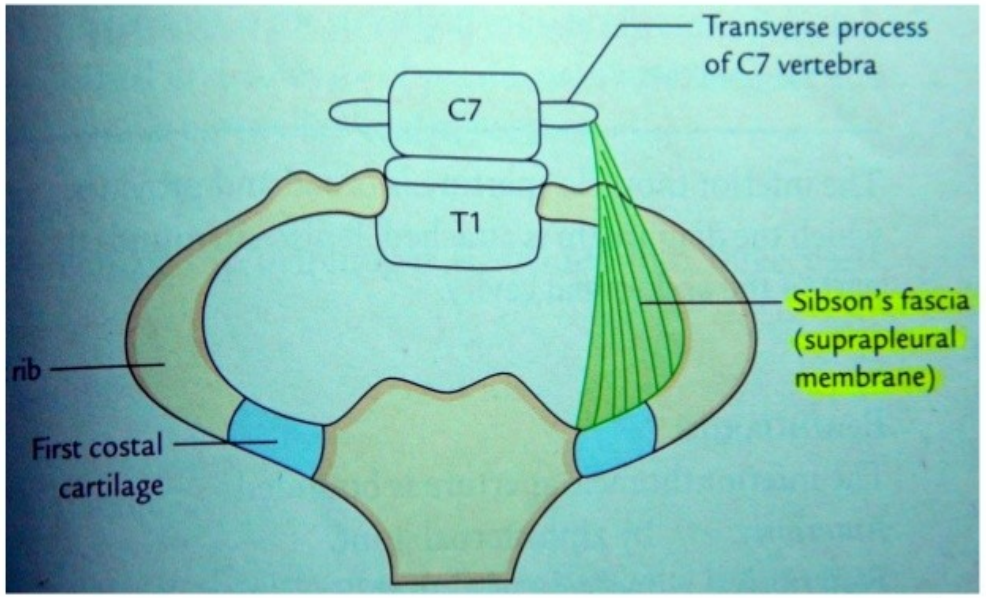


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Suprapleural membrane(sibson's fascia)

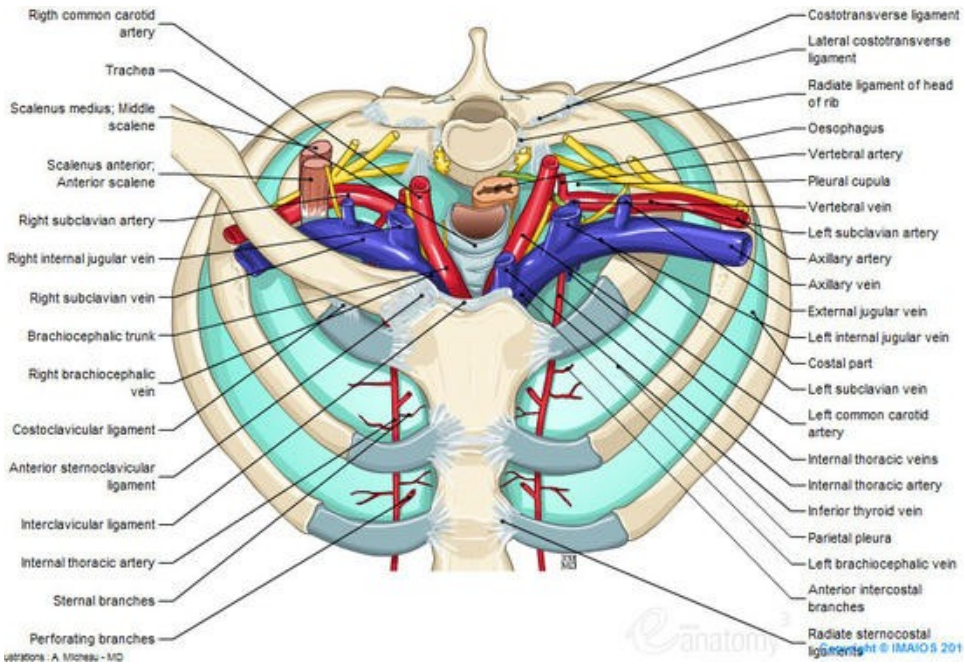
- Partly separates thorax from neck.
- Attachment
- Apex
  - To tip of transverse process of C7 vertebra
- Base to
  - Inner border of 1st rib & its cartilage.

Suprapleural membrane/ Sibson's fascia



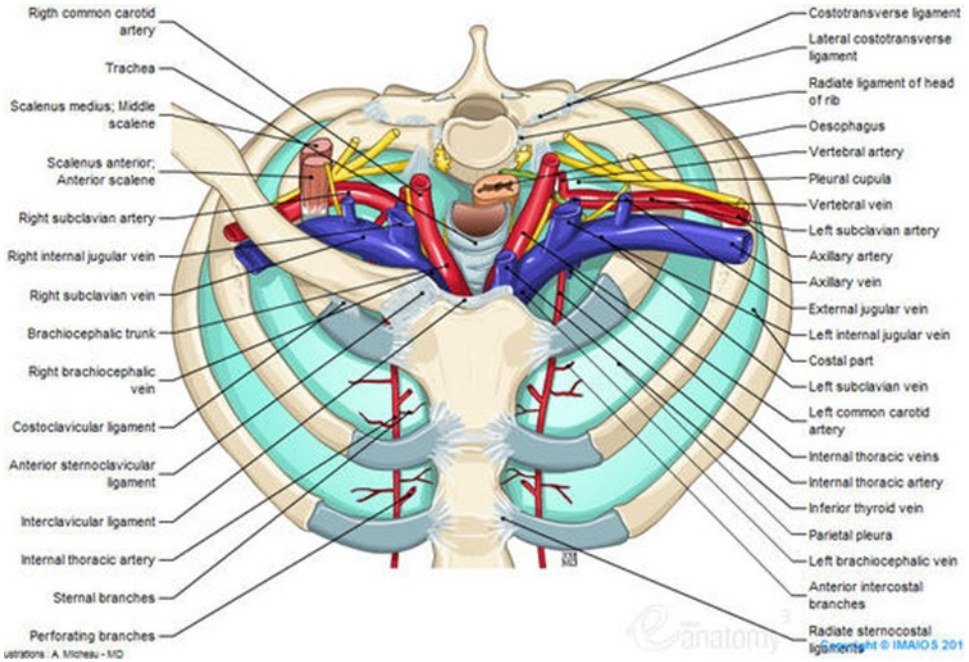
Structures passing through inlet

- Viscera
  - Oesophagus.
  - Trachea.
  - Apices of lung.
- Large vessels.
  - Right bracheocephali artery &vein
  - Left common carotid artery.
  - Left subclavian artery,
  - Left brachiocephalic vein.



Structures passing through inlet

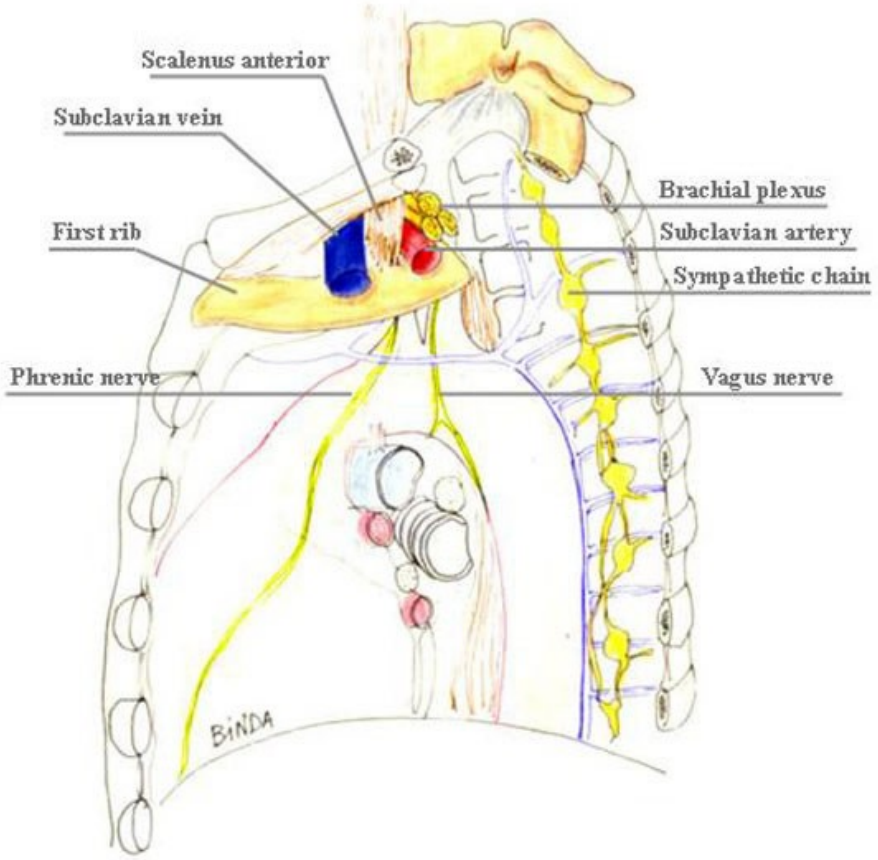
- Small vessels
  - Right & left internal thoracic arteries.
  - Right & left superior intercostal arteries.
  - Right &left 1st posterior intercostal veins.
  - Inferior thyroid veins.



Structures passing through inlet

- Nerves.

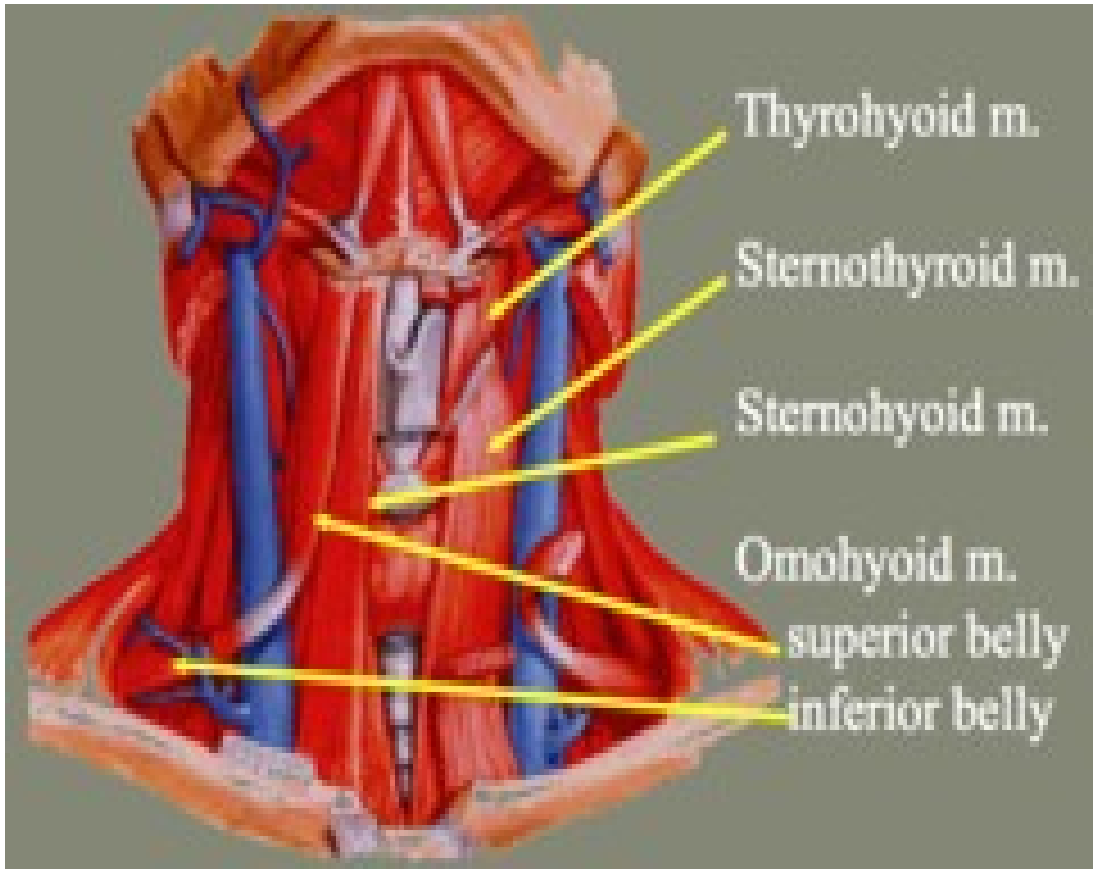
1. Right & left phrenic nerves.
2. Right & left vagus
3. Right & left sympathetic trunks
4. Right & left 1st thoracic nerves.



Structures passing through inlet

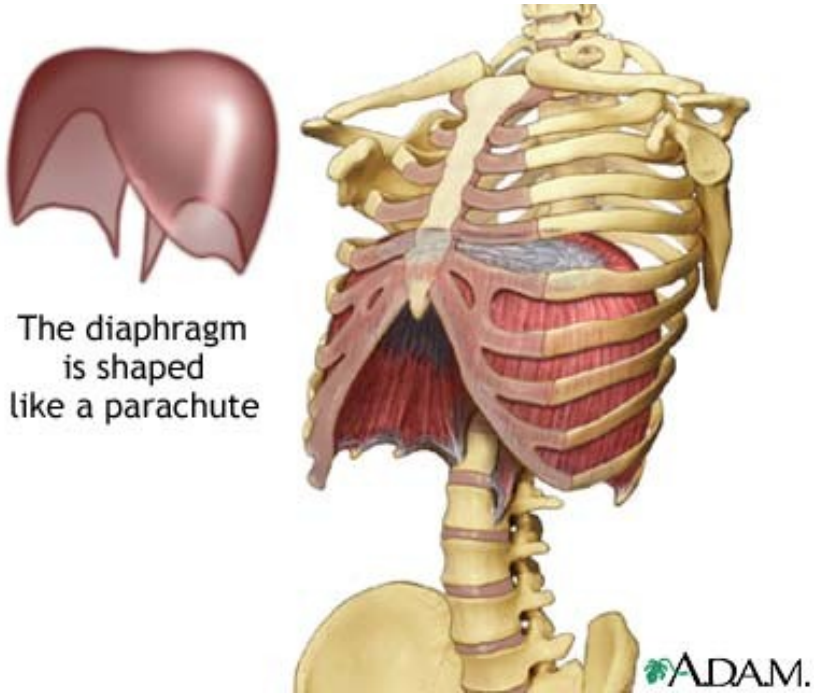
- Muscles.

1. Sternohyoid.
2. Sternothyroid
3. Longus colli.



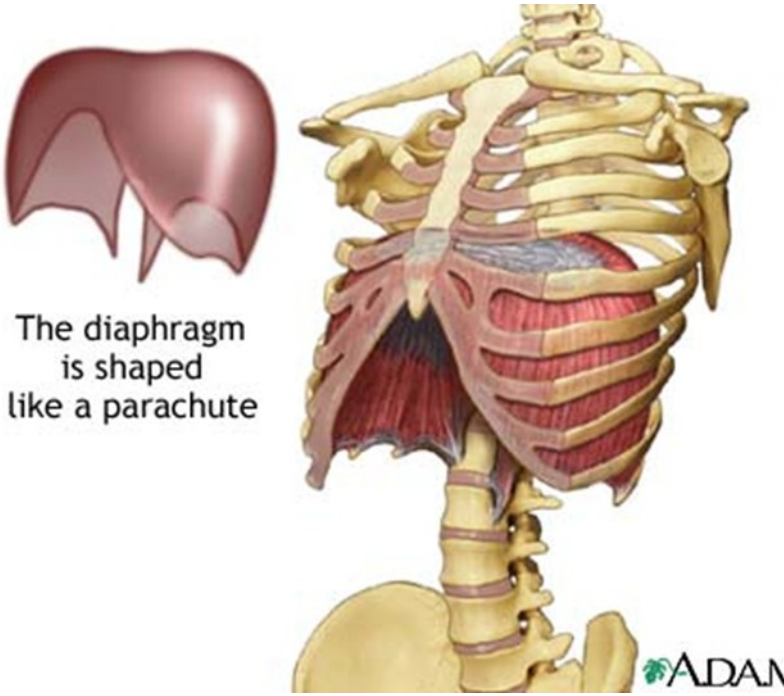
Thoracic outlet

1. Is broad end of thorax.
2. Surrounds the upper part of abdominal cavity.
3. Separated from abdomen by diaphragm.



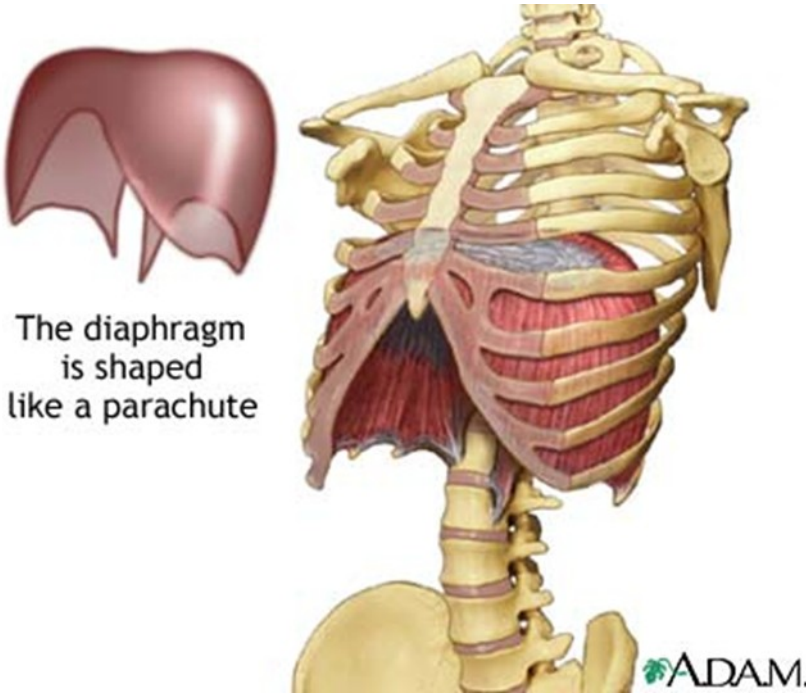
Boundaries of thoracic outlet

- Anteriorly
  - Infrasternal angle
- Posteriorly.
  - Inferior surface of the body of 12th vertebra.
- On each side.
  - Costal margin formed by the cartillages of 7th -12th rib.



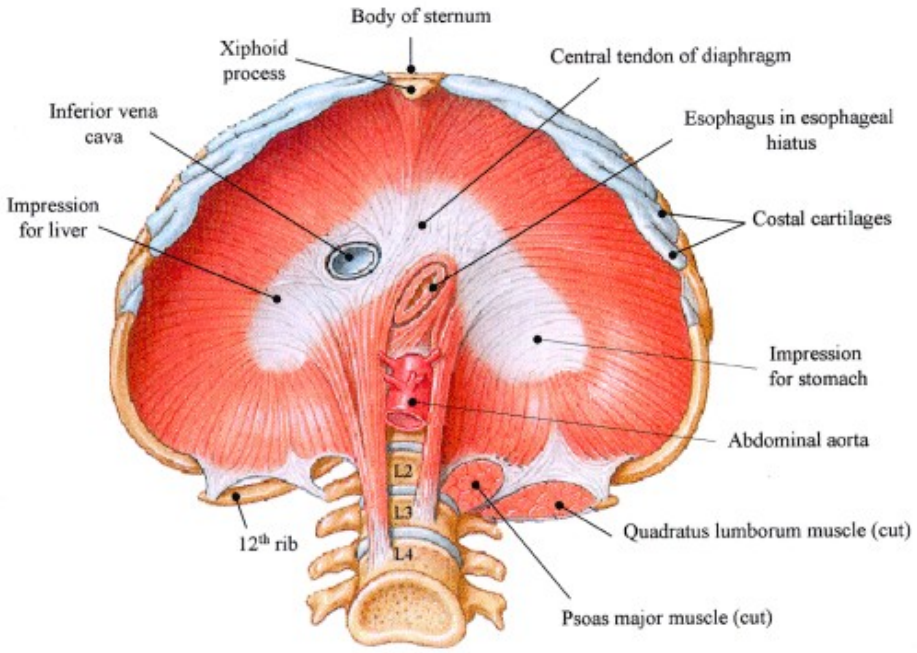
Diaphragm of the outlet of thorax

- Musculotendinous partition that close the outlet.
- Called thoracoabdominal diaphragm.



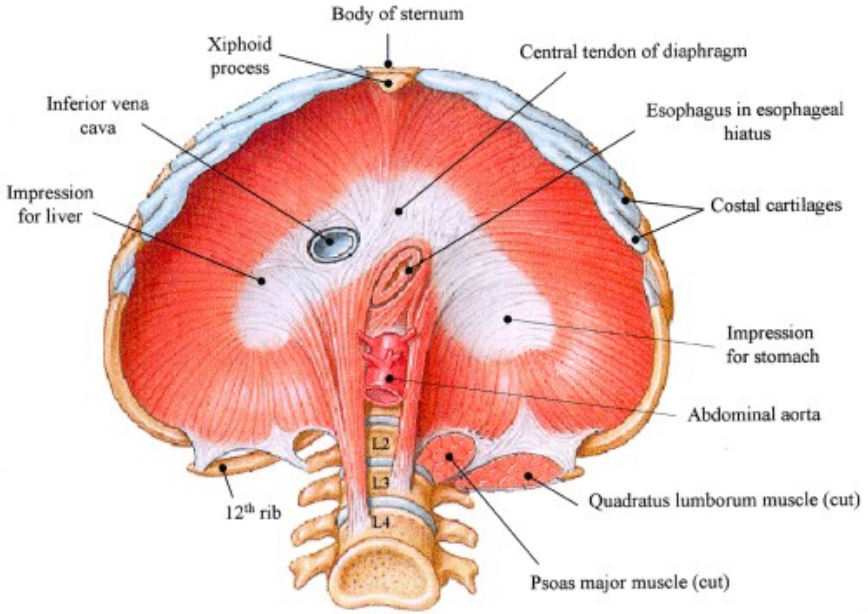
Large openings in diaphragm

- Aortic opening.
- Is Osseo aponeurotic.
- Lies at lower border of the T12 vertebra.
- Transmit
  - Aorta.
  - Thoracic duct.
  - Azygos vein.



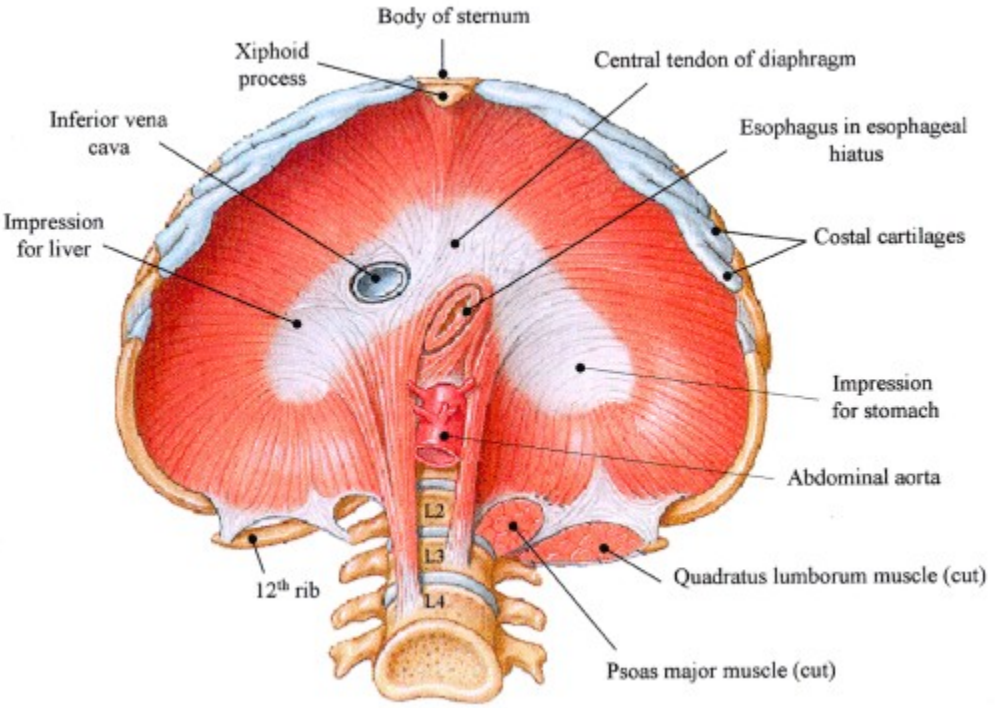
Large openings in diaphragm

- Oesophageal opening.
- Lies in the muscular part.
- At 10th thoracic vertebra.
- Transmit.
  - Oesophagus.
  - Gastric (vagus) nerves.
  - Oesophageal branches of the left gastric artery.



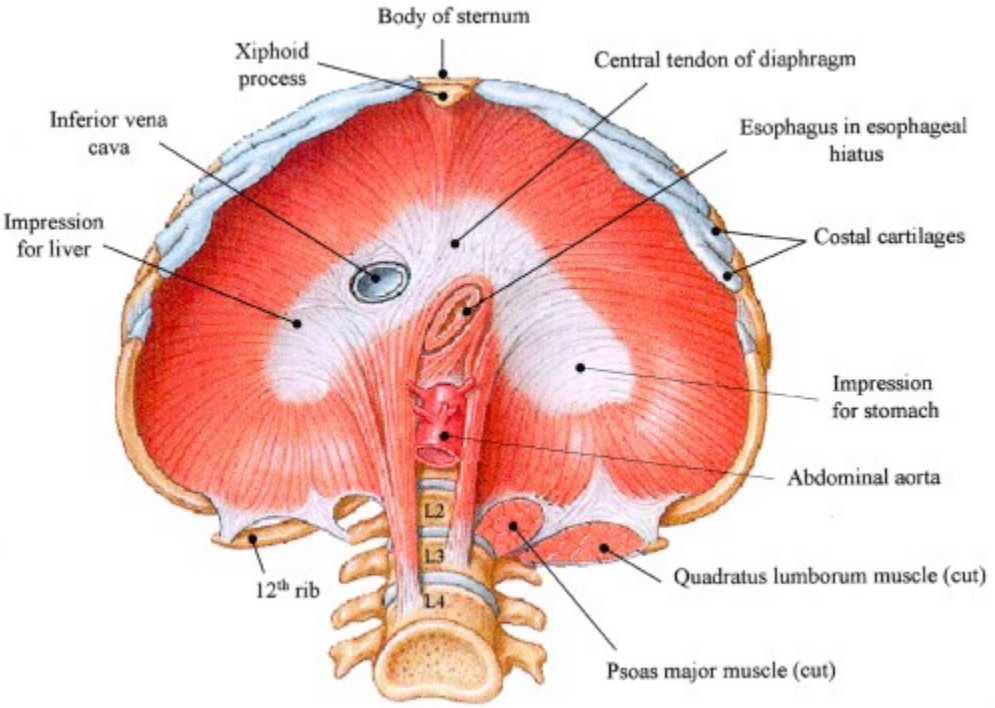
Large openings in diaphragm

- Venacaval opening.
- Lies in the central tendon.
- At 8th thoracic vertebra.
- Transmit
  - Inferior venacava.
  - Branches of the right phrenic nerve.



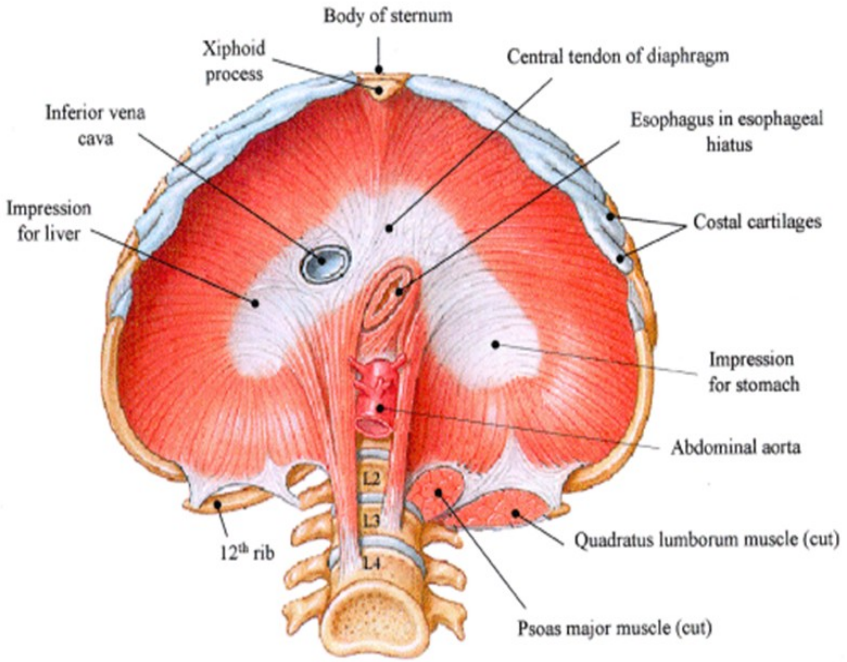
Small openings in diaphragm

1. Each crus
    1. Greater & lesser splanchic nerves.
    2. Hemiazygos vein in left crus.
2. Behind medial arcuate line.
    1. Sympathetic chain
3. Behind lateral arcuate ligament
    1. Subcostal nerve.



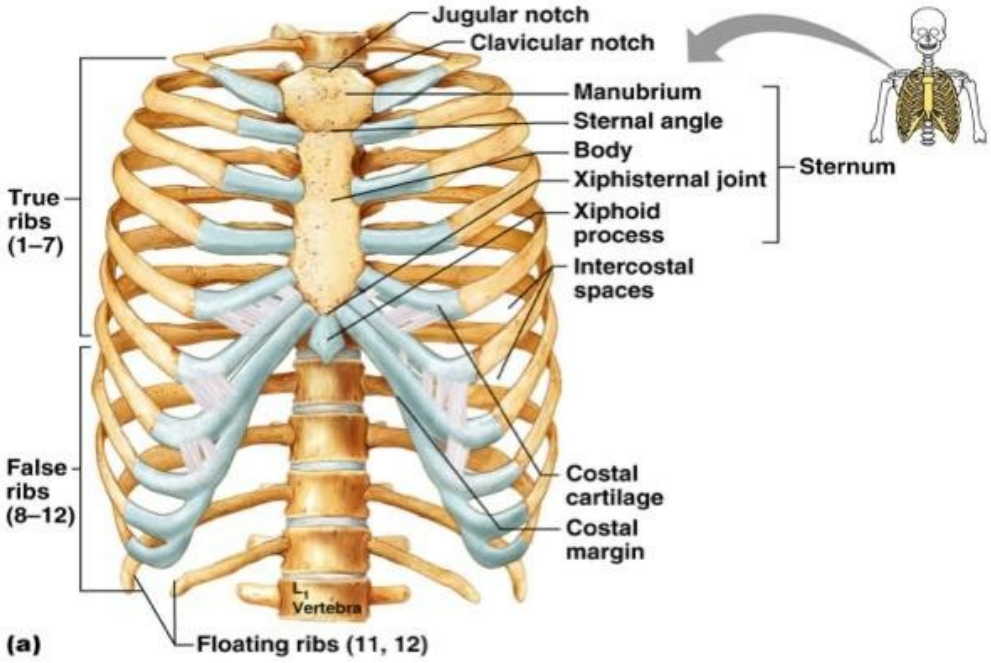
Small openings in diaphragm

1. b/w xiphoid & costal (7th costal cartilage) origin of diaphragm space is known as Larry's space or foramen of Morgagni pass
    1. Superior epigastric vessels & lymphatics.
2. At 9th costal cartilages
    1. Musculophrenic nerve.
3. Minute aperture's in central tendon
    1. Several small veins.



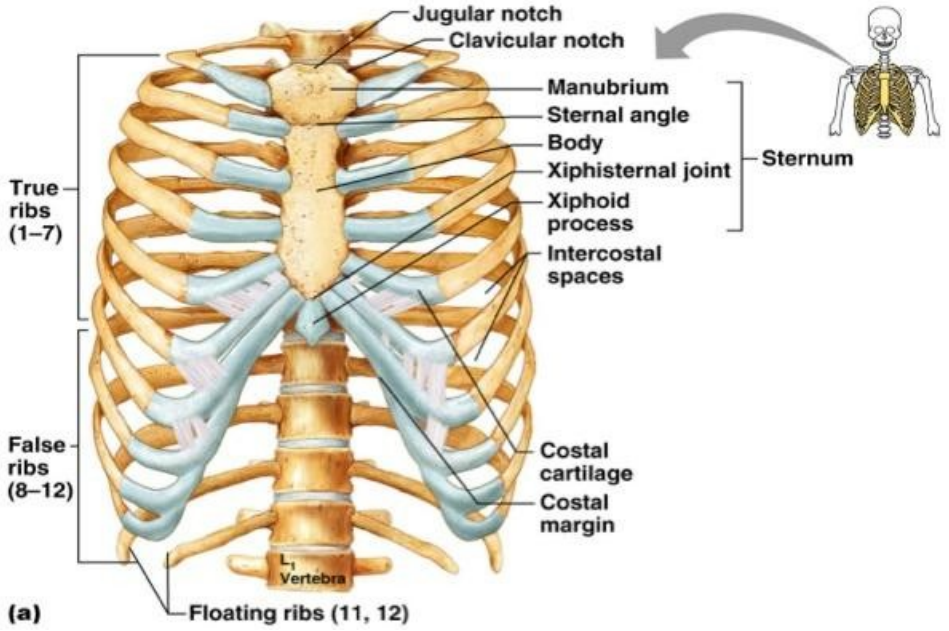
Surface land marks of the thorax

1. Suprasternal or jugular notch lies at lower border of T2 vertebra.
2. Sternal angle of Loius.
    1. Marks the manubrosternal joint
    2. Felt 5cm below the suprasternal notch.
    3. Lies at the level of the 2nd costal cartilage anteriorly & disc b/w 4th & 5th thoracic vertebra posteriorly.



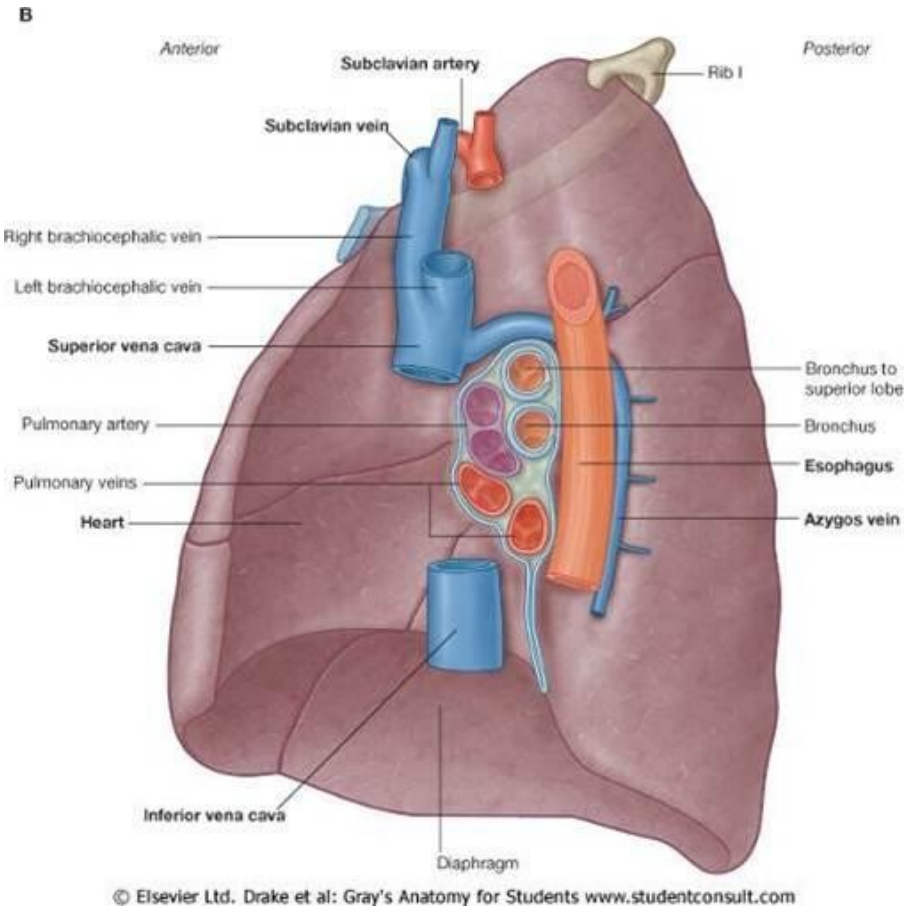
Angle of Louis
important landmarks

1. Ribs are counted.
2. Plane b/w superior & inferior mediastinum.
3. Ascending aorta ends.
4. Arch of the aorta begins & ends.
5. Descending aorta begins.
6. Trachea divides into principal bronchi.



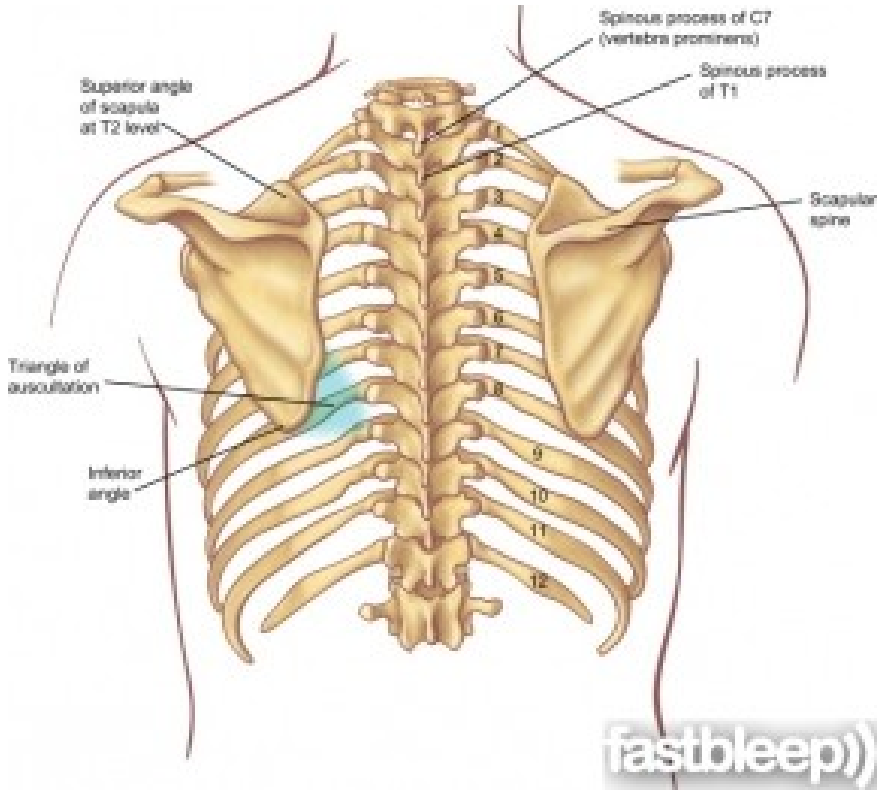
Angle of Louis
important landmarks

- 7. azygos vein arches over root of the right lung & opens into superior venacava.
- 8. pulmonary trunk divides into two pulmonary arteries just below.
- Thoracic duct crosses from right to left side.
- Marks the upper limit of the base of the heart.
- Cardiac plexuses are situated.



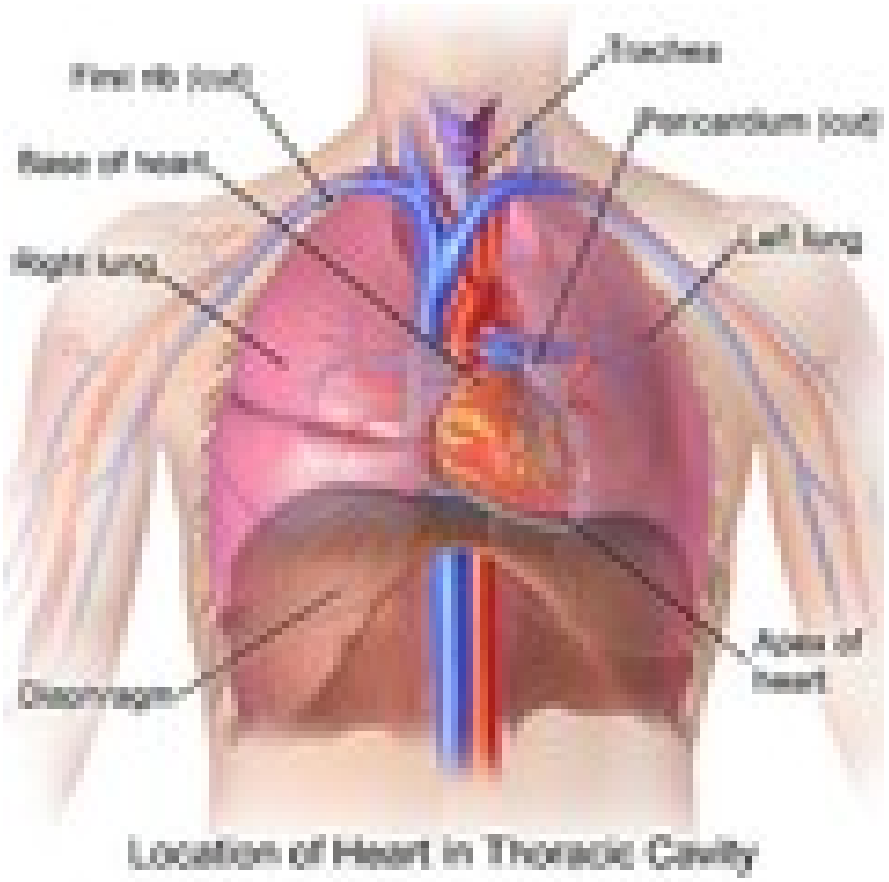
Important landmarks

- Costal cartilages.
  - 2nd at sternal angle.
  - 7th at infrasternal angle or xiphisternal joint.
- Ribs
  - Scapula overlies at 2nd -7th ribs .
  - 10th rib lies at 3rd lumbar vertebra.
- Thoracic vertebrae
  - Spine of T3 lies at spine of scapula.
  - Spine of 7th lies at inferior angle.



Thoracic cavity

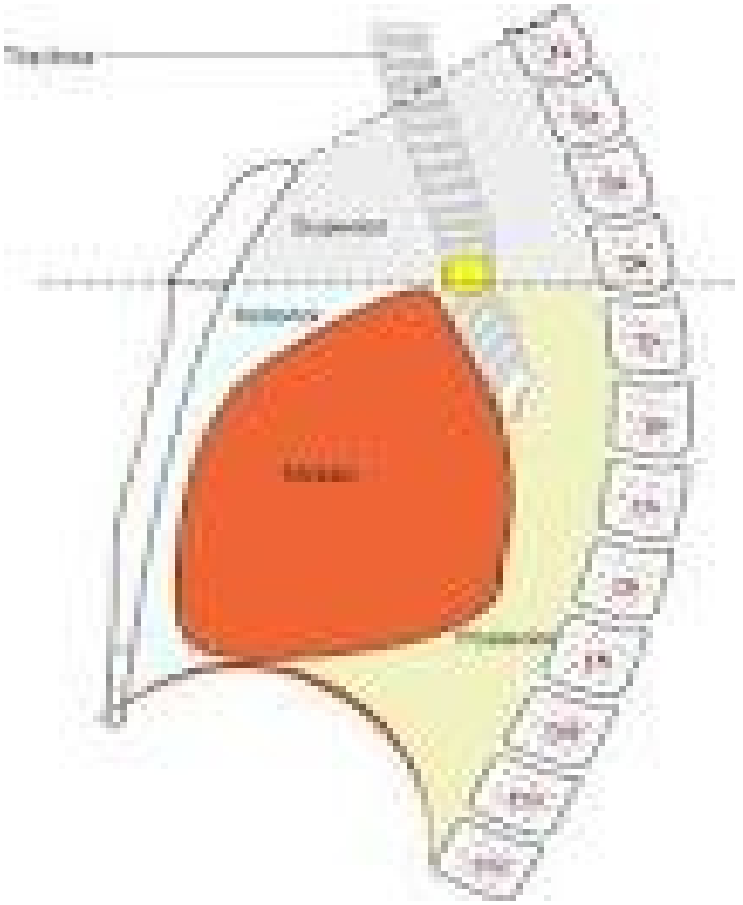
1. Filled laterally by lungs.
2. Each lung lies in pleural cavity.
3. Space b/w the pleural cavities occupying the center of thoracic cavity is mediastinum.



Division of mediastinum

- By plane passing from sternal angle to 4th thoracic vertebra into

1. Superior mediastinum
2. Inferior mediastinum divided by pericardium into
    1. Anterior
    2. Middle
    3. Posterior.



THANKYOU